

3_10xMouse_PBMC_comparison_wSTAR

August 31, 2026

```
[1]: library(Seurat)
library(ggplot2)
library(ggbeeswarm)
library(pheatmap)
library(ggpubr)
getwd()
dir.create("figures_10xMouse_PBMC")
dir.create("data")

colorAge = c("old"="#7a646a",
             "young"="#88bce7")

colorTools = c("SoloTE"="#A4DD9B",
               "Stellarscope"="#4f5d93",
               "STARsolo"="#f0df93")

dataset_id <- "10xMouse_PBMC"
```

Loading required package: SeuratObject

Loading required package: sp

‘SeuratObject’ was built under R 4.3.3 but the current version is 4.4.1; it is recommended that you reinstall ‘SeuratObject’ as the ABI for R may have changed

‘SeuratObject’ was built with package ‘Matrix’ 1.6.4 but the current version is 1.7.5; it is recommended that you reinstall ‘SeuratObject’ as the ABI for ‘Matrix’ may have changed

Attaching package: ‘SeuratObject’

The following objects are masked from ‘package:base’:

intersect, t

```
'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'
```

```
Warning message in dir.create("figures_10xMouse_PBMC"):
```

```
"'figures_10xMouse_PBMC' already exists"
```

```
Warning message in dir.create("data"):
```

```
"'data' already exists"
```

```
[2]: theme_paper <- function(base_size = 17, base_family = "") {  
  theme_minimal(base_size = base_size, base_family = base_family) +  
    theme(  
      plot.title = element_text(face = "bold", size = base_size + 2, hjust = 0.  
↪5),  
      axis.title = element_text(size = base_size),  
      axis.text = element_text(size = base_size * 0.9),  
      legend.title = element_text(size = base_size),  
      legend.text = element_text(size = base_size * 0.9),  
      panel.grid.major = element_blank(),  
      panel.grid.minor = element_blank(),  
      plot.margin = margin(10, 10, 10, 10)  
    )  
}  
theme_set(theme_paper())
```

1 Pre

```
[28]: # Import Stellarscope object  
objTE_stellarscope <- readRDS(paste0("data_",dataset_id,"/  
↪stellarscope_",dataset_id,"_seuratObj.RDS"))  
# Import SoloTE object  
objTE_soloTE <- readRDS(paste0("data_",dataset_id,"/  
↪soloTE_",dataset_id,"_seuratObj.RDS"))  
# Import STARsolo object  
objTE_STARsolo <- readRDS(paste0("data_",dataset_id,"/  
↪STARsolo_",dataset_id,"_seuratObj.RDS"))
```

```
[29]: objTE_stellarscope  
objTE_soloTE  
objTE_STARsolo
```

An object of class Seurat

4123 features across 4787 samples within 1 assay

Active assay: RNA (4123 features, 4000 variable features)

3 layers present: counts, data, scale.data

2 dimensional reductions calculated: pca, umap

An object of class Seurat

5430 features across 4787 samples within 1 assay
 Active assay: RNA (5430 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap

An object of class Seurat
 3182 features across 4790 samples within 1 assay
 Active assay: RNA (3182 features, 3182 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap

```
[30]: conversionTable <- read.table("/mnt/TEdata/TEbenchmarking/Rscripts/annotation/
      ↪annotation_mm10_conversion_withAge.tsv")
      head(conversionTable)
```

A data.frame: 6 × 16

	chr	start	end	strand	sw_score	locusID	family	subfam
	<chr>	<int>	<int>	<chr>	<int>	<chr>	<chr>	<chr>
1	chr1	3000001	3002128	-	12955	L1_Mus3_dup24	L1	L1-Mu
2	chr1	3003153	3003994	-	1216	L1Md_F_dup2	L1	L1Md-
3	chr1	3003994	3004054	-	234	L1_Mus3_dup25	L1	L1-Mu
4	chr1	3004041	3004206	+	3685	L1_Rod_dup1	L1	L1-Rod
5	chr1	3004271	3005001	+	3685	L1_Rod_dup2	L1	L1-Rod
6	chr1	3005002	3005439	+	1280	L1_Rod_dup3	L1	L1-Rod

```
[31]: # get TEs in common between Stellarscope and SoloTE

stellarscopeTEs <- Features(objTE_stellarscope)

table(stellarscopeTEs %in% conversionTable$stellarscopeID)

soloTEs <- Features(objTE_soloTE)
# remove "SoloTE" from the name of the TEs and substitute "/" and "_" with "-"
soloTEs <- gsub("\\|", "-", soloTEs)
soloTEs <- gsub("\\_", "-", soloTEs)
soloTEs <- gsub("\\?", "", soloTEs)

table(soloTEs %in% conversionTable$soloteID)

TEsInStellarscope_SoloTE <- intersect(
  ↪conversionTable$locusID[conversionTable$stellarscopeID %in% stellarscopeTEs],
  ↪conversionTable$locusID[conversionTable$soloteID %in%
  ↪soloTEs] )
length(TEsInStellarscope_SoloTE)
```

TRUE
 4123

FALSE TRUE

5 5425

3006

```
[32]: # get TEs in common between Stellarscope and STARsolo

stellarscopeTEs <- Features(objTE_stellarscope)

table(stellarscopeTEs %in% conversionTable$stellarscopeID)

STARsoloTEs <- Features(objTE_STARsolo)

table(STARsoloTEs %in% conversionTable$stellarscopeID)

TEsInStellarscope_STARsolo <- intersect(
  ↪conversionTable$locusID[conversionTable$stellarscopeID %in% stellarscopeTEs],
  ↪conversionTable$locusID[conversionTable$stellarscopeID
  ↪%in% STARsoloTEs] )
length(TEsInStellarscope_STARsolo)
```

TRUE

4123

TRUE

3182

1360

```
[33]: STARsoloTEs <- Features(objTE_STARsolo)
table(STARsoloTEs %in% conversionTable$stellarscopeID)

TEsInAll <- intersect( TEsInStellarscope_SoloTE,
  ↪conversionTable$locusID[conversionTable$stellarscopeID
  ↪%in% STARsoloTEs] )

length(TEsInAll)
```

TRUE

3182

901

```
[34]: commonCellsAll <- intersect(intersect(Cells(objTE_stellarscope),
  ↪Cells(objTE_soloTE)), Cells(objTE_STARsolo))
length(commonCellsAll)
```

4787

```
[35]: ## locus id of common TEs
tes_soloTE_conv <- conversionTable$locusID[match(Features(objTE_soloTE),
↳conversionTable$soloteID)]
tes_stellarscope_conv <-
↳conversionTable$locusID[match(Features(objTE_stellarscope),
↳conversionTable$stellarscopeID)]
tes_starsolo_conv <- conversionTable$locusID[match(Features(objTE_STARsolo),
↳conversionTable$stellarscopeID)]

commonTEs_conv <- intersect(intersect(tes_soloTE_conv, tes_stellarscope_conv),
↳tes_starsolo_conv)
commonTEs_conv <- commonTEs_conv[!is.na(commonTEs_conv)]
length(commonTEs_conv)
```

901

```
[36]: # ALL common TEs
# revert to original name
common_soloTE <- conversionTable$soloteID[match(commonTEs_conv,
↳conversionTable$locusID)]
common_stellarscope <- conversionTable$stellarscopeID[match(commonTEs_conv,
↳conversionTable$locusID)]
common_STARsolo <- conversionTable$stellarscopeID[match(commonTEs_conv,
↳conversionTable$locusID)]

# extract matrix of common TEs
stellarscopeMatSub <- GetAssayData(objTE_stellarscope[common_stellarscope,
↳commonCellsAll], layer = "data")
setdiff(common_stellarscope, rownames(stellarscopeMatSub))
stellarscopeMatSub <- stellarscopeMatSub[common_stellarscope, commonCellsAll] #
↳to reorder

STARsoloMatSub <- GetAssayData(objTE_STARsolo[common_stellarscope,
↳commonCellsAll], layer = "data")
STARsoloMatSub <- STARsoloMatSub[common_stellarscope, commonCellsAll] # to
↳reorder

soloTEMatSub <- GetAssayData(objTE_soloTE[common_soloTE, commonCellsAll], layer
↳= "data")
soloTEMatSub <- soloTEMatSub[common_soloTE, commonCellsAll] # to reorder

identical(colnames(stellarscopeMatSub), colnames(soloTEMatSub))
identical(colnames(STARsoloMatSub), colnames(soloTEMatSub))

matSubs <- list()
```

```
matSubs[["Stellarscope"]] <- stellarscopeMatSub
matSubs[["SoloTE"]] <- soloTEMatSub
matSubs[["STARsolo"]] <- STARsoloMatSub

gc()
```

TRUE

TRUE

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2×6 of type dbl	Ncells	22620502	1208.1	45891062	2450.9	33960431	1813.7
	Vcells	246680443	1882.1	445544962	3399.3	444741507	3393.2

```
[37]: # matrices with common cells and TEs
mat_stellarscope <- as.matrix(stellarscopeMatSub)

mat_soloTE <- as.matrix(soloTEMatSub)

mat_STARsolo <- as.matrix(STARsoloMatSub)

gc()
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2×6 of type dbl	Ncells	22620226	1208.1	45891062	2450.9	33960431	1813.7
	Vcells	259619112	1980.8	445544962	3399.3	444741507	3393.2

2 Correlations

```
[38]: avgExprCommonTEs <- rowMeans(mat_stellarscope)
# compute average expression in the cells where each TE is expressed
sumExprCommonTEs <- rowSums(mat_stellarscope)
nCellsExprTEs <- rowSums(mat_stellarscope>0)
avgExprPerExprCell <- sumExprCommonTEs/nCellsExprTEs

TEcorrsCommonTEs <- list()

# compute spearman correlation of stellarscope and soloTE counts
TEcorrsCommonTEs[["Stellarscope_SoloTE"]] <- sapply(1:nrow(mat_stellarscope),
  FUN = function(i){
    cor(mat_stellarscope[i,], mat_soloTE[i,], method="spearman")
  })

# compute spearman correlation of stellarscope and STARsolo counts
TEcorrsCommonTEs[["Stellarscope_STARsolo"]] <- sapply(1:nrow(mat_stellarscope),
  FUN = function(i){
    cor(mat_stellarscope[i,], mat_STARsolo[i,], method="spearman")
  })
```

```

})

# compute spearman correlation of stellarscope and STARSolo counts
TEcorrsCommonTEs[["STARSolo_SoloTE"]] <- apply(1:nrow(mat_STARSolo), FUN =
  ↪function(i){
    cor(mat_STARSolo[i,], mat_soloTE[i,], method="spearman")
  })

```

```

[39]: summary(TEcorrsCommonTEs[["Stellarscope_SoloTE"]])
summary(TEcorrsCommonTEs[["Stellarscope_STARSolo"]])
summary(TEcorrsCommonTEs[["STARSolo_SoloTE"]])

```

```

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.02509 0.90163 0.95861 0.91709 0.98165 0.99997

```

```

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.09149 0.74711 0.87553 0.83264 0.97051 0.99969

```

```

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.03275 0.67081 0.82495 0.76820 0.93002 0.99765

```

3 Checkpoint

```

[3]: #save.image(paste0("workspaces/afterCorrelations_wSTAR_thr5percCells_",
  ↪dataset_id, ".RData"))
load(paste0("workspaces/afterCorrelations_wSTAR_thr5percCells_", dataset_id, ".
  ↪RData"))
rm(mat_stellarscope)
rm(mat_soloTE)
rm(mat_STARSolo)
gc()

```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2 × 6 of type dbl	Ncells	22095283	1180.1	37915637	2025.0	22109562	1180.8
	Vcells	248122264	1893.1	373324491	2848.3	269544806	2056.5

4 Post

```

[41]: options(repr.plot.width=5, repr.plot.height=4)

for(combo in names(TEcorrsCommonTEs)){

  tools <- unlist(strsplit(combo, "_"))
  show(
    ggplot() + aes(x=TEcorrsCommonTEs[[combo]]) +
      geom_density(color="#99918F", fill=alpha("#99918F",0.
  ↪5), linewidth = 1) +

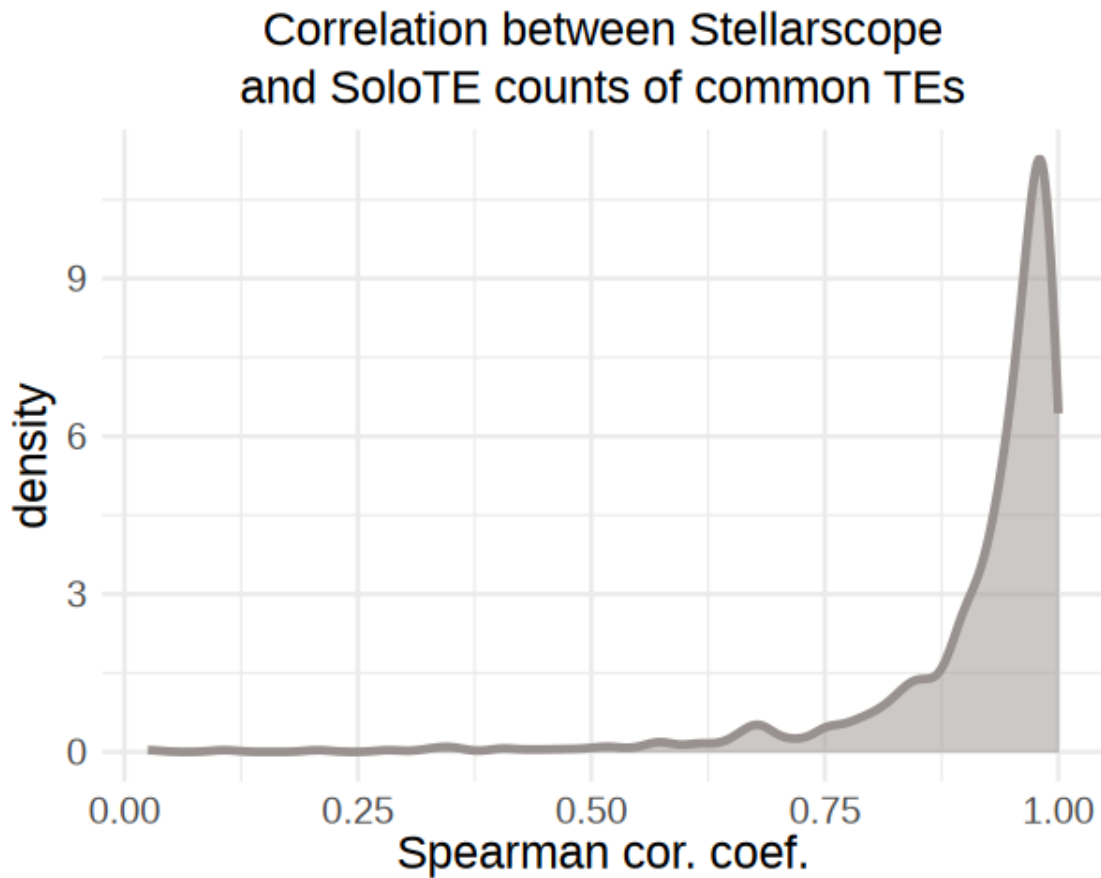
```

```

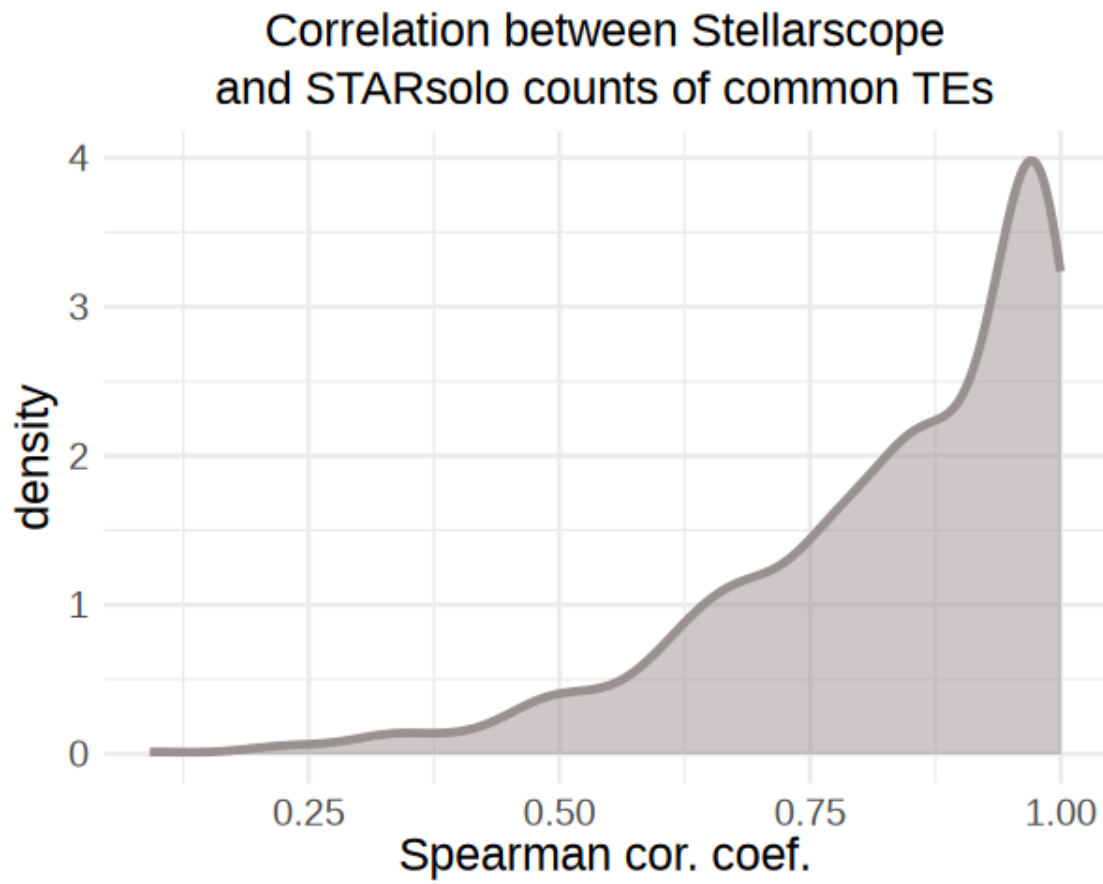
    ggtitle(paste0("Correlation between ", tools[1]),
             subtitle = paste0("and ", tools[2], " counts of_",
    ↪common TEs")) +
    xlab("Spearman cor. coef.") +
    theme_minimal() +
    theme(text=element_text(size=14.5),
           plot.title = element_text(size=14.5, hjust=0.
    ↪5),
           plot.subtitle = element_text(size=14.5, hjust=0.
    ↪5))
  )
  ggsave(paste0("figures_", dataset_id, "/"
    ↪spearmanCorrelation_density_commonTEs_", combo, ".pdf"), device = "pdf")
}

```

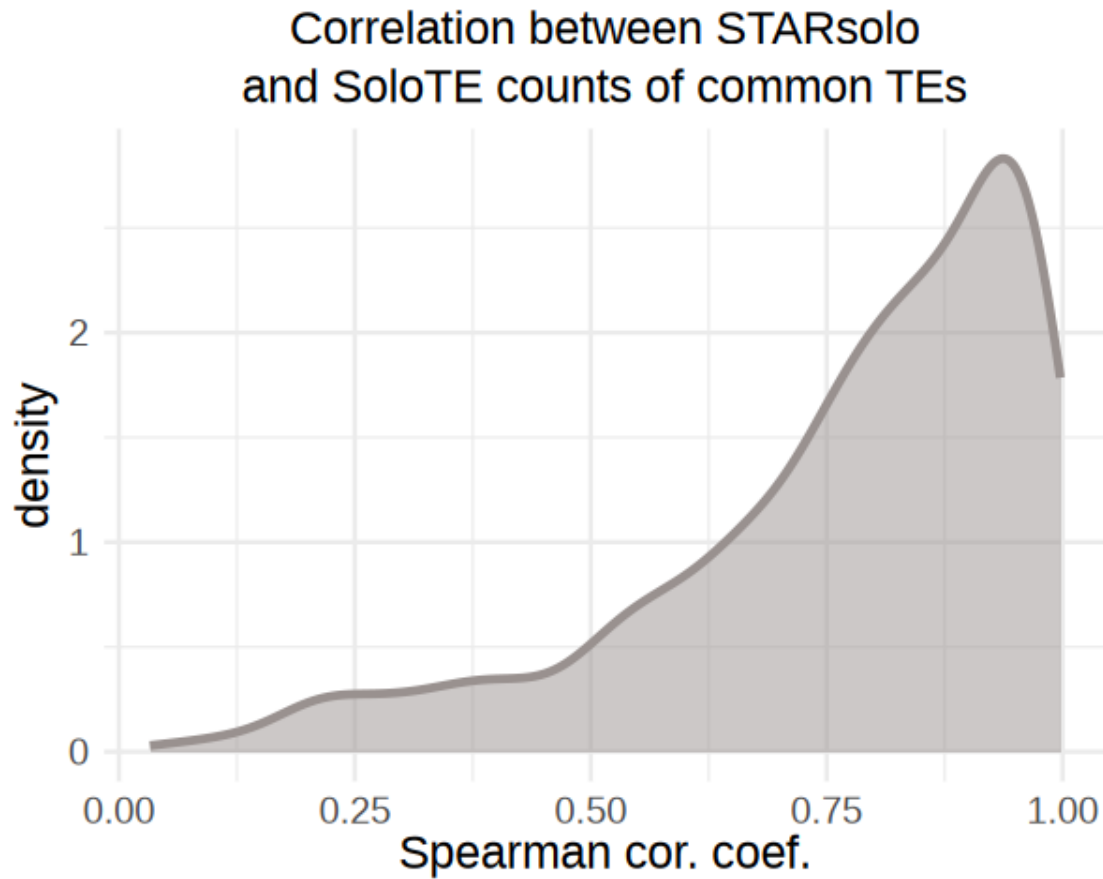
Saving 6.67 x 6.67 in image



Saving 6.67 x 6.67 in image



Saving 6.67 x 6.67 in image



```
[42]: # define age breaks for bins
breaks <- c(0, 2, 5, 10, 15, 25, 40, 60, Inf)

summary(conversionTable$mya)

conversionTable$age_bin <- cut(
  conversionTable$mya,
  breaks = breaks,
  labels = c("0-2", "2-5", "5-10", "10-15", "15-25", "25-40", "40-60", "60+"),
  right = FALSE,
  include.lowest = TRUE
)

table(conversionTable$age_bin)

table(conversionTable$age_bin[is.na(conversionTable$mya)])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
0.00	18.87	30.84	29.32	39.84	88.60	7134

0-2	2-5	5-10	10-15	15-25	25-40	40-60	60+
25622	111376	312875	265885	645109	1440271	901650	17530

0-2	2-5	5-10	10-15	15-25	25-40	40-60	60+
0	0	0	0	0	0	0	0

4.1 Correlation by age

```
[43]: options(repr.plot.width=10, repr.plot.height=6)
library(ggbeeswarm)

dfsCorrelationAge <- list()

for(combo in names(TEcorrsCommonTEs)){
  tools <- unlist(strsplit(combo, "_"))

  df <- cbind(rownames(matSubs[[tools[1]]]), TEcorrsCommonTEs[[combo]])
  df <- as.data.frame(df)
  colnames(df) <- c("V1", "TEcorrsCommonTEs")

  df$ageClass <- conversionTable$ageClass[match(df$V1,
↪conversionTable$stellarscopeID)]
  df$age_bin <- conversionTable$age_bin[match(df$V1,
↪conversionTable$stellarscopeID)]
  df$mya <- conversionTable$mya[match(df$V1, conversionTable$stellarscopeID)]
  df$TEcorrsCommonTEs <- as.numeric(df$TEcorrsCommonTEs)
  cor(df$mya, df$TEcorrsCommonTEs, method="spearman")

  show( ggplot(df, aes(x=age_bin, y=TEcorrsCommonTEs, fill=age_bin)) +
        geom_violin(alpha = 0.8) +
        stat_summary(fun.data = function(x) data.frame(y = max(x), label =
↪length(x)),
        geom = "text", vjust = -0.5, size = 5, color = "black") +
        theme_paper() +
        #scale_fill_manual(values=colorAge) +
        guides(fill = "none") +
        theme(text=element_text(size=18)) +
        xlab("Age bins (mya)") +
        ggtitle(combo)
      )

  show( ggplot(df, aes(x=age_bin, y=TEcorrsCommonTEs, color=age_bin)) +
        geom_beeswarm(alpha = 0.7, cex = 0.2, size=0.5, width=0.5) +
        #geom_boxplot(outlier.shape = NA, width = 0.3, color = "gray40",
↪fill=NA, alpha = 0.3) +
```

```

    stat_summary(
      fun = median,
      geom = "crossbar",
      width = 0.3,
      color = "grey30",
      linewidth = 0.5
    ) +
      stat_summary(fun.data = function(x) data.frame(y = max(x), label =
↪length(x)),
                  geom = "text", vjust = -0.5, size = 5, color = "black") +
      theme_paper() +
      #scale_fill_manual(values=colorAge) +
      guides(color = "none") +
      theme(text=element_text(size=18)) +
      xlab("Age bins (mya)") +
      ylab("Spearman cor. coef.") +
      ggtitle(combo)
  )
  ggsave(paste0("figures_", dataset_id, "/correlation_byAge_Bin_", combo, ".pdf"),
↪width=8, height=6)

  show( ggplot(df, aes(x=ageClass, y=TEcorrsCommonTEs, fill=ageClass)) +
        geom_violin(alpha = 0.5) +
        stat_summary(fun.data = function(x) data.frame(y = max(x), label =
↪length(x)),
                    geom = "text", vjust = -0.5, size = 5, color = "black") +
        theme_pubclean() +
        scale_fill_manual(values=colorAge) +
        theme(text=element_text(size=18)) +
        xlab("Age class") +
        ggtitle(combo)
  )

  # divide by correlation quartiles
  df$quartile <- "4"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.75, na.
↪rm=TRUE)] <- "3"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.5, na.
↪rm=TRUE)] <- "2"
  df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.25, na.
↪rm=TRUE)] <- "1"

  show(ggplot(df, aes(x=quartile, y=mya)) +
        geom_violin(fill=alpha("#99918F",0.5)) +
        stat_summary(fun.data = function(x) data.frame(y = max(x), label =
↪length(x)),

```

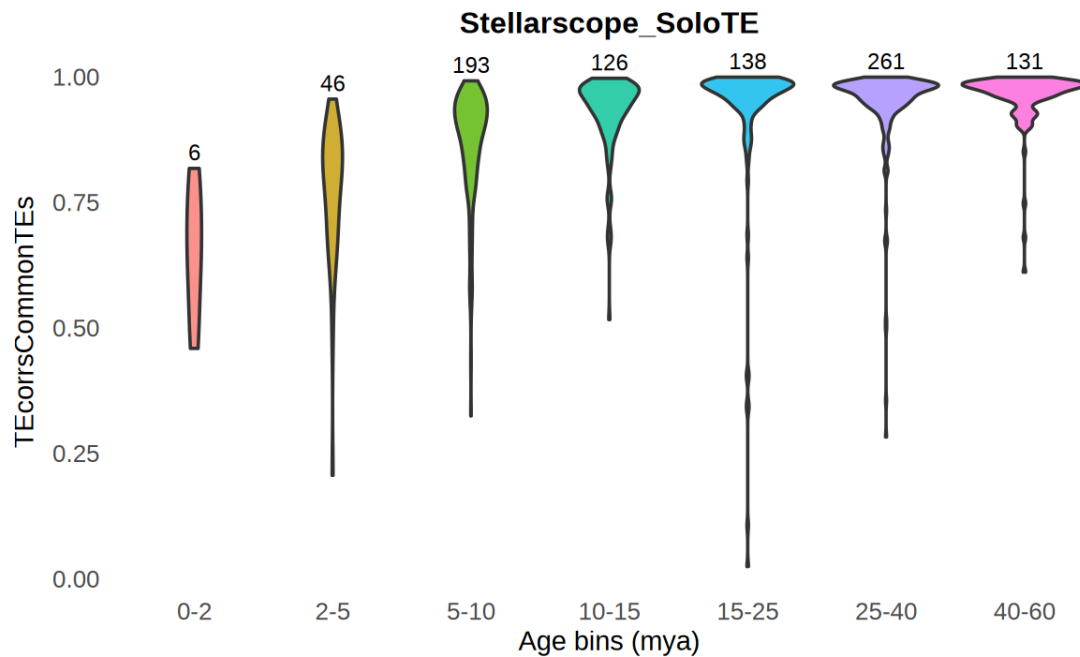
```

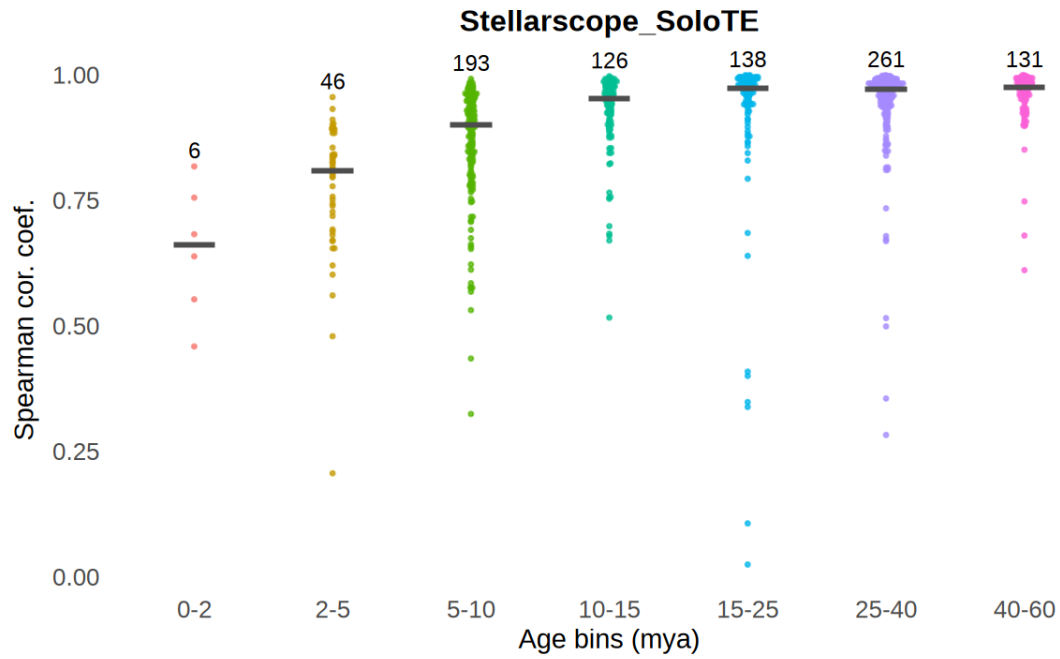
      geom = "text", vjust = -0.5, size = 5) +
    theme_pubclean() +
    theme(text=element_text(size=18)) +
    scale_y_continuous(transform = "log") +
    xlab("Correlation quartile") +
    ggtitle(combo)
  )

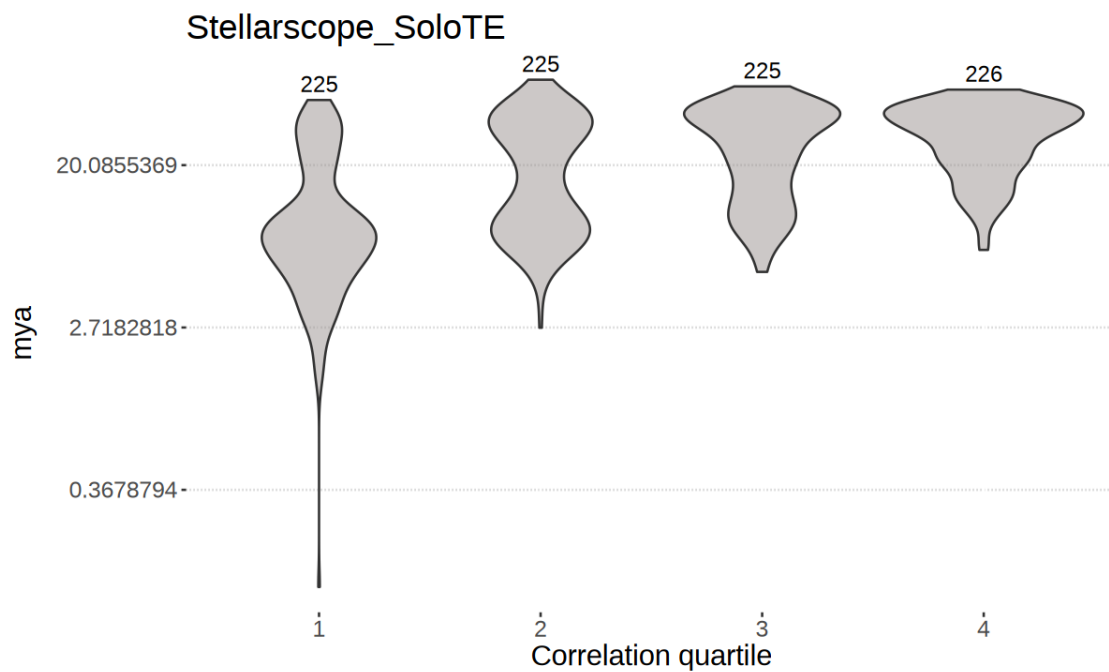
  dfsCorrelationAge[[combo]] <- df
}

```

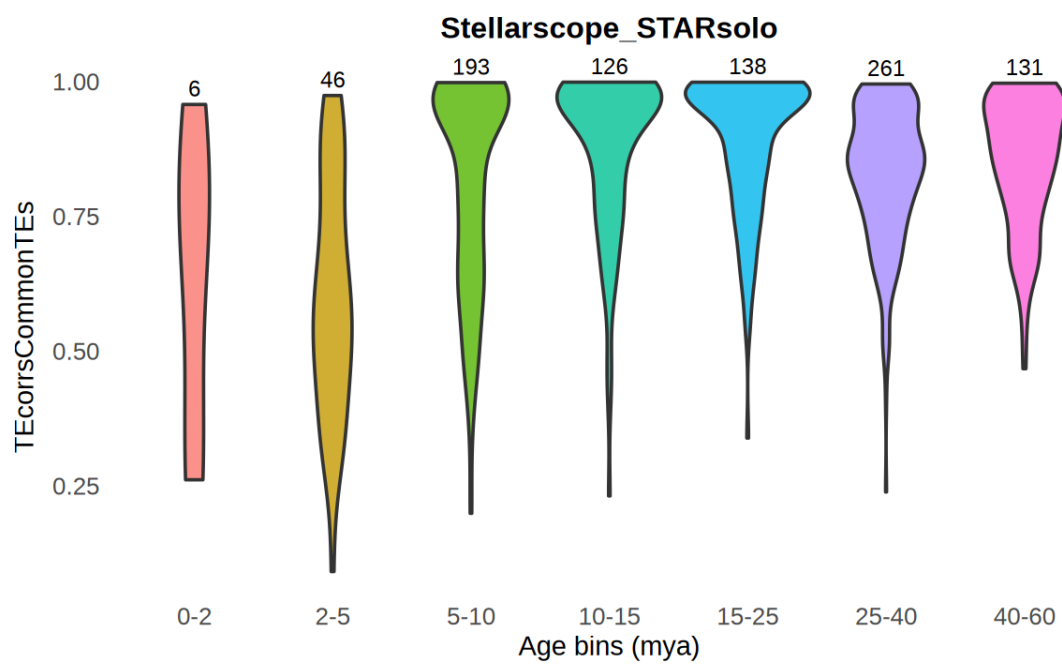
Warning message in `geom_beeswarm(alpha = 0.7, cex = 0.2, size = 0.5, width = 0.5)`:
 "Ignoring unknown parameters: `width`"

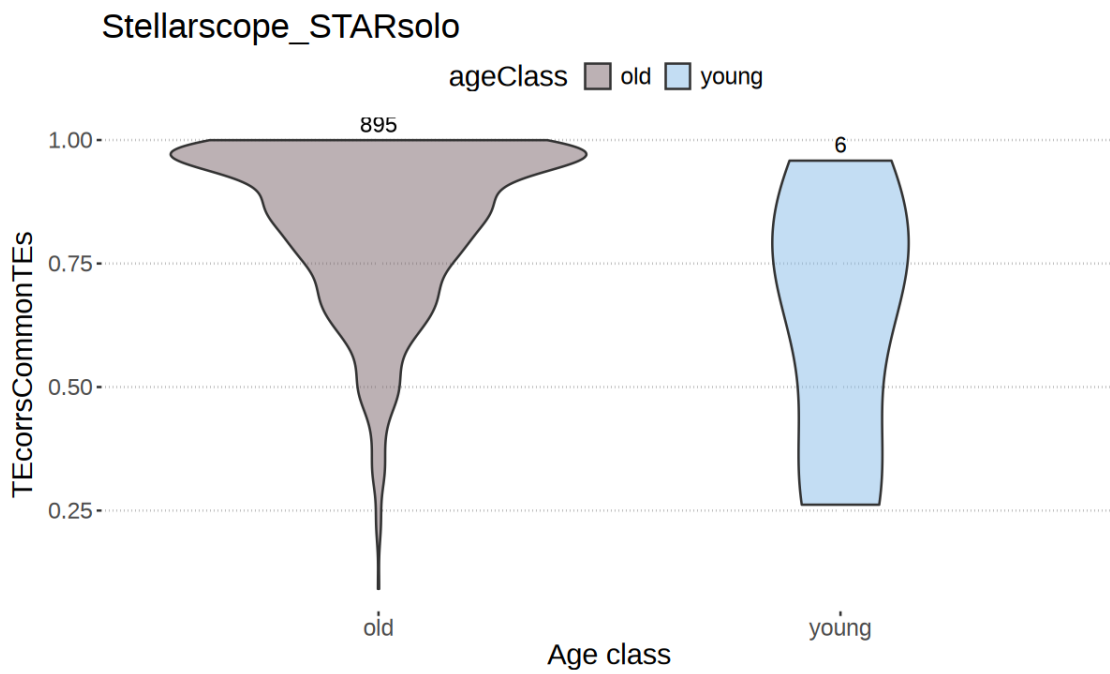
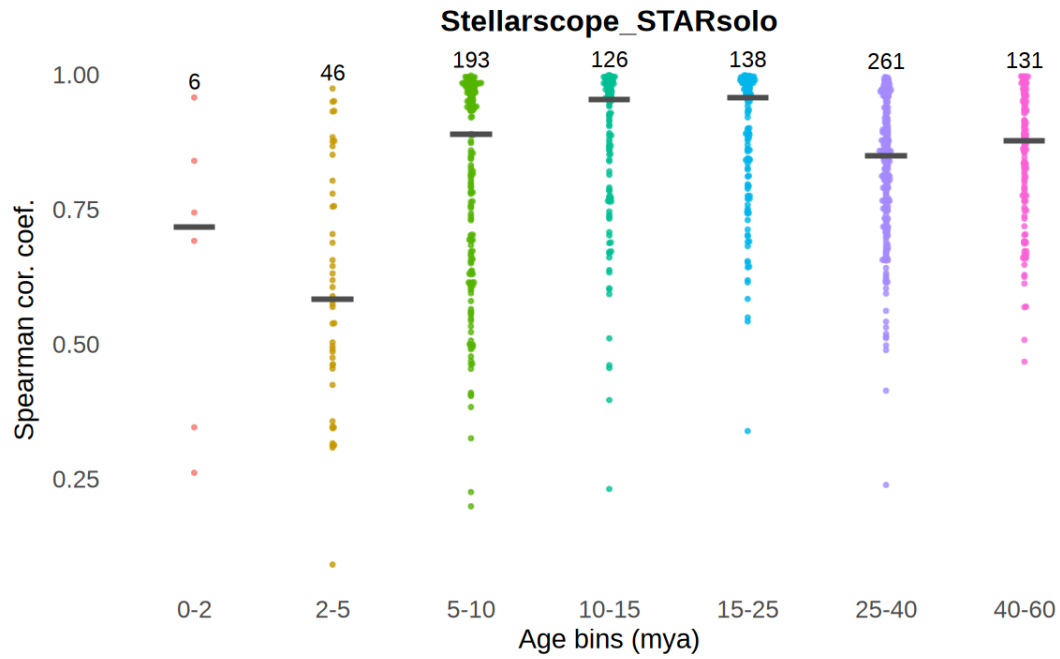


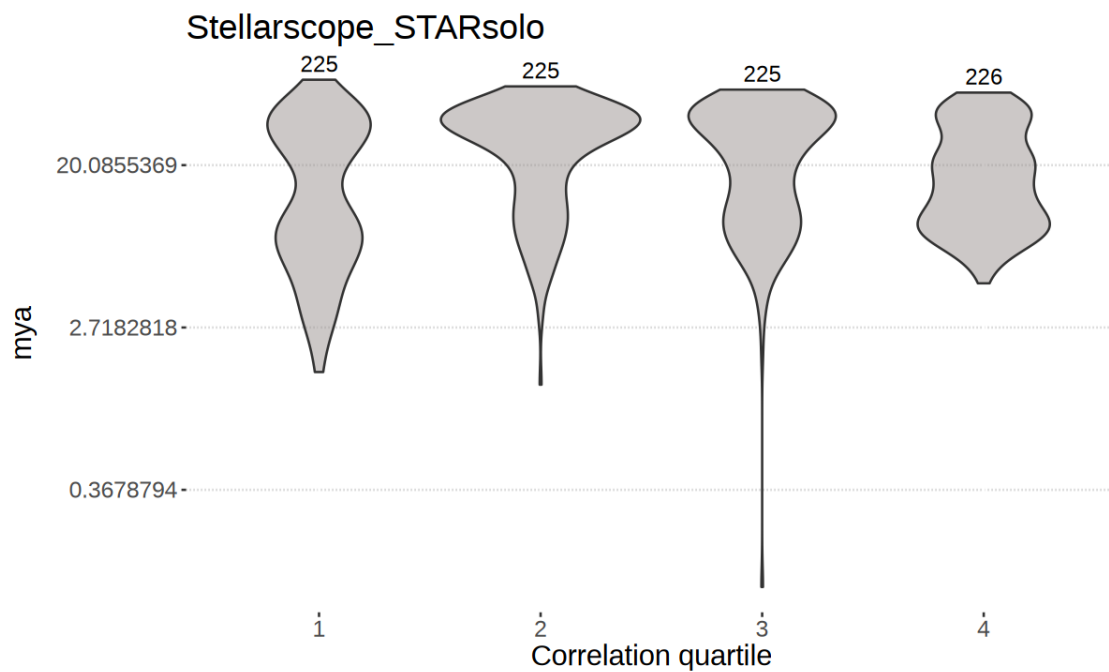




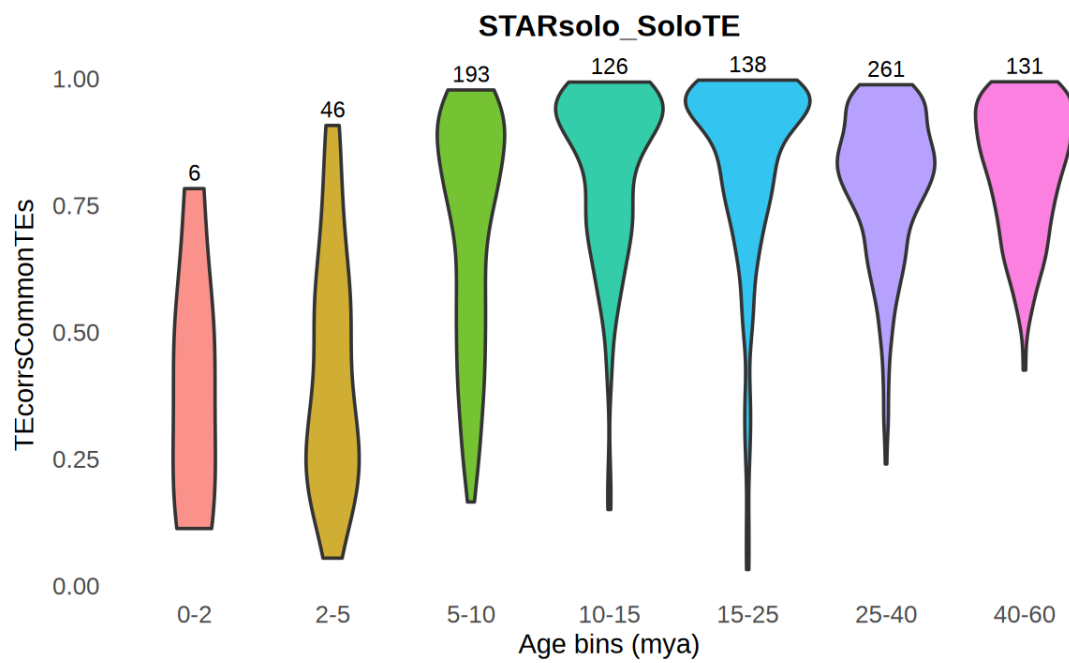
Warning message in `geom_beeswarm(alpha = 0.7, cex = 0.2, size = 0.5, width = 0.5)`:
 "Ignoring unknown parameters: `width`"

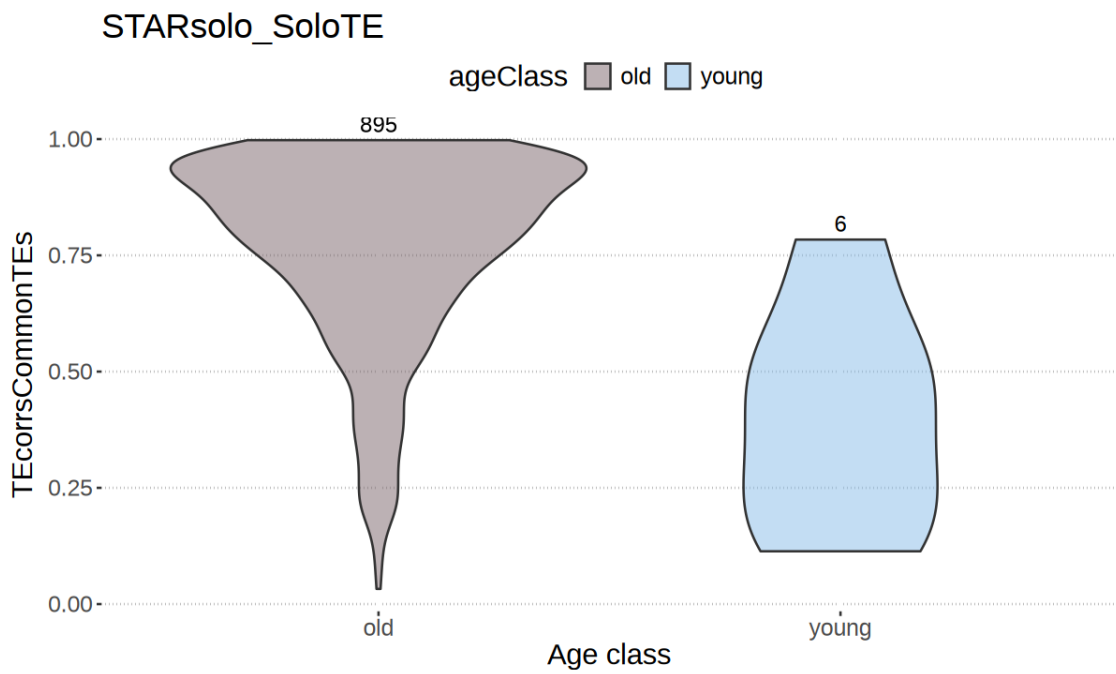
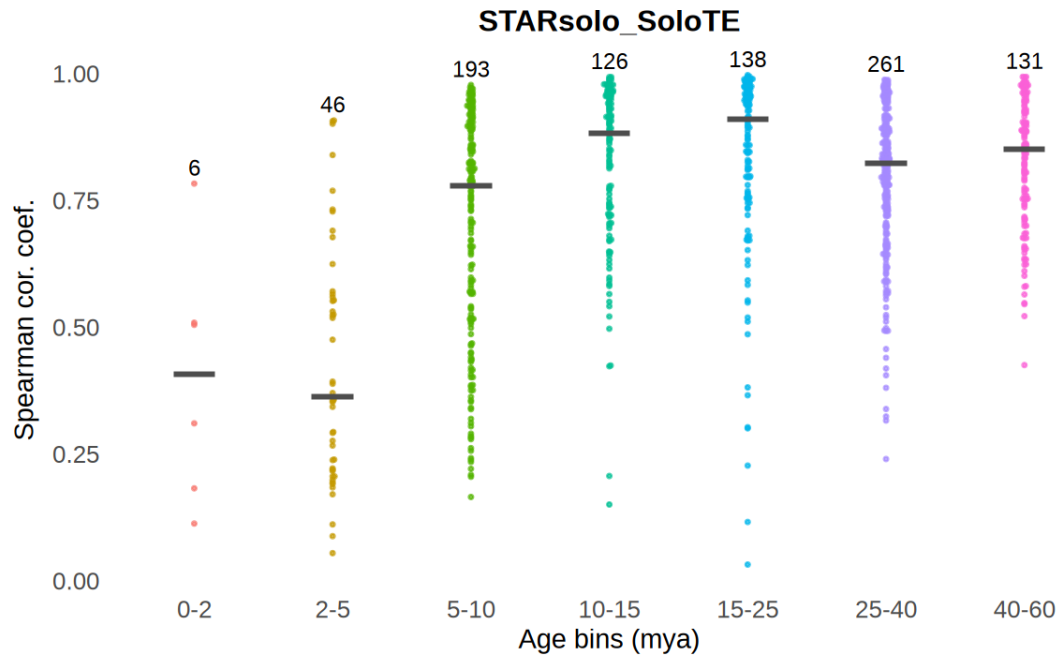






Warning message in `geom_beeswarm(alpha = 0.7, cex = 0.2, size = 0.5, width = 0.5)`:
 "Ignoring unknown parameters: ``width``"







```
[44]: options(repr.plot.width=5, repr.plot.height=6)

dfsCorrelationAge <- list()

for(combo in names(TEcorrsCommonTEs)){
  tools <- unlist(strsplit(combo, "_"))

  df <- cbind(rownames(matSubs[[tools[1]]]), TEcorrsCommonTEs[[combo]])
  df <- as.data.frame(df)
  colnames(df) <- c("V1", "TEcorrsCommonTEs")

  df$ageClass <- conversionTable$ageClass[match(df$V1,
↪conversionTable$stellarscopeID)]
  df$mya <- conversionTable$mya[match(df$V1, conversionTable$stellarscopeID)]
  df$TEcorrsCommonTEs <- as.numeric(df$TEcorrsCommonTEs)
  cor(df$mya, df$TEcorrsCommonTEs, method="spearman")

  options(repr.plot.width=5, repr.plot.height=6)

  show( ggplot(df, aes(x=ageClass, y=TEcorrsCommonTEs, fill=ageClass)) +
    geom_violin(jitter=TRUE, alpha = 0.5) +
    theme_pubclean() +
    scale_fill_manual(values=colorAge) +
    theme(text=element_text(size=18)) +
    xlab("Age class") +
```

```

        ggtitle(combo)
    )

# with jitter
# show(ggplot(df, aes(x=ageClass, y=TEcorrsCommonTEs, fill=ageClass)) +
#   geom_violin(jitter=TRUE, alpha = 0.5) +
#   geom_jitter(position = position_jitter(seed = 1, width = 0.2), size=0.
↪1, alpha=0.2) +
#   theme_pubclean() +
#   scale_fill_manual(values=colorAge) +
#   theme(text=element_text(size=18)) +
#   xlab("Age class") +
#   ggtitle(combo)
# )

# divide by correlation quartiles
df$quartile <- "4"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.75, na.
↪rm=TRUE)] <- "3"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.5, na.
↪rm=TRUE)] <- "2"
df$quartile[df$TEcorrsCommonTEs < quantile(df$TEcorrsCommonTEs, 0.25, na.
↪rm=TRUE)] <- "1"

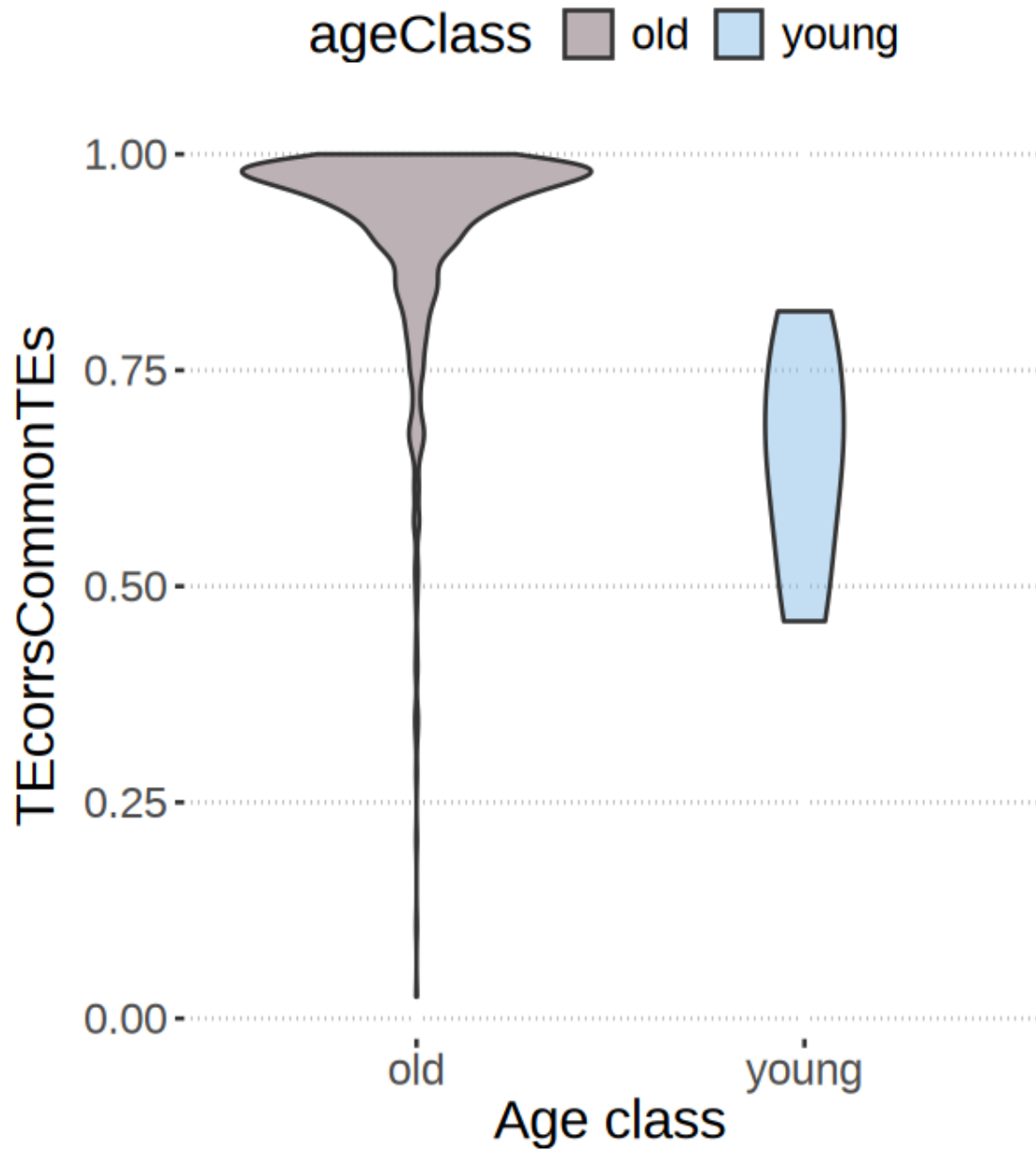
show(ggplot(df, aes(x=quartile, y=mya)) +
  geom_violin(fill=alpha("#99918F",0.5)) +
  theme_pubclean() +
  theme(text=element_text(size=18)) +
  scale_y_continuous(transform = "log") +
  xlab("Correlation quartile") +
  ggtitle(combo)
)

dfsCorrelationAge[[combo]] <- df
}

```

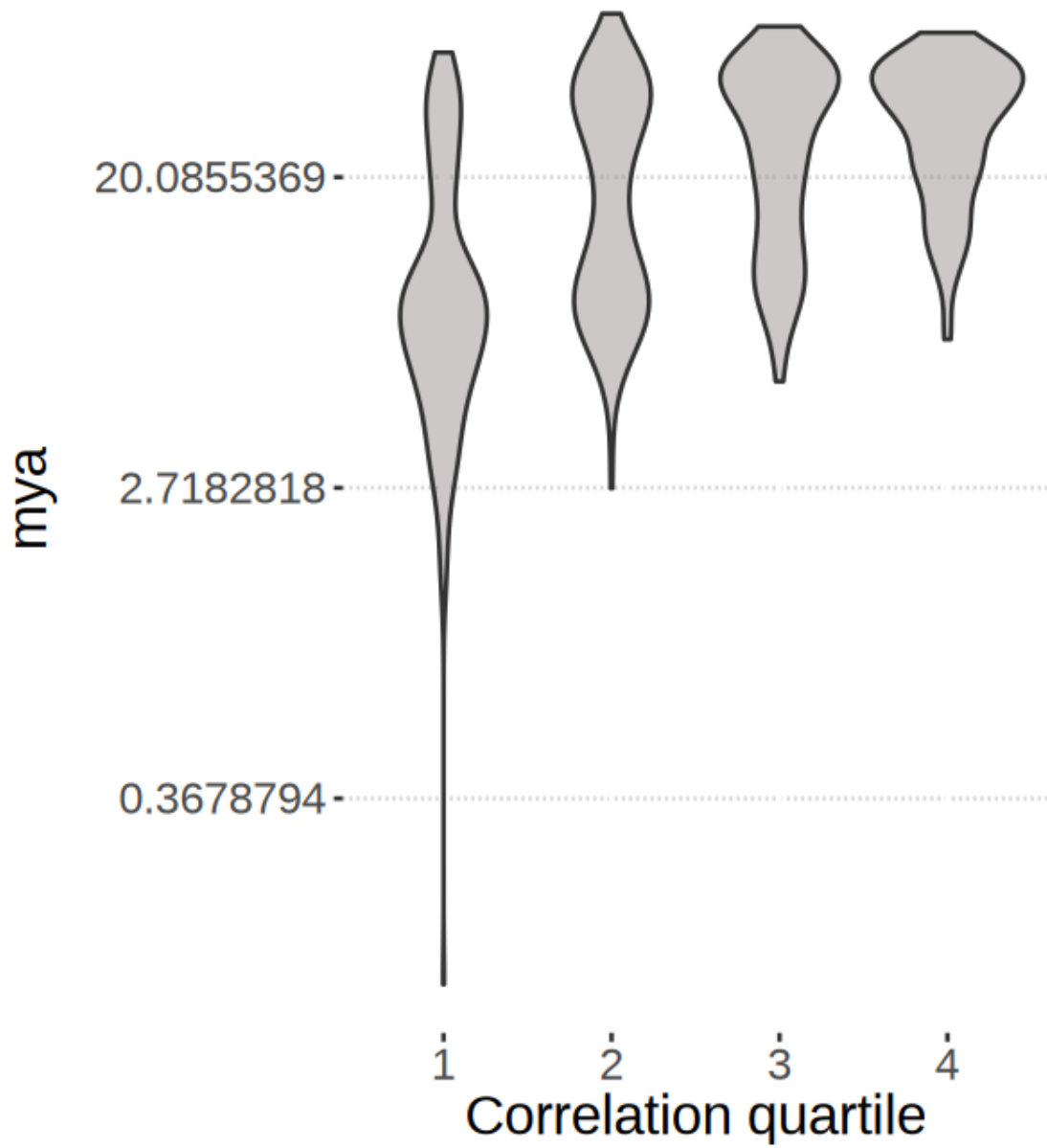
Warning message in geom_violin(jitter = TRUE, alpha = 0.5):
 "Ignoring unknown parameters: `jitter`"

Stellarscope_SoloTE

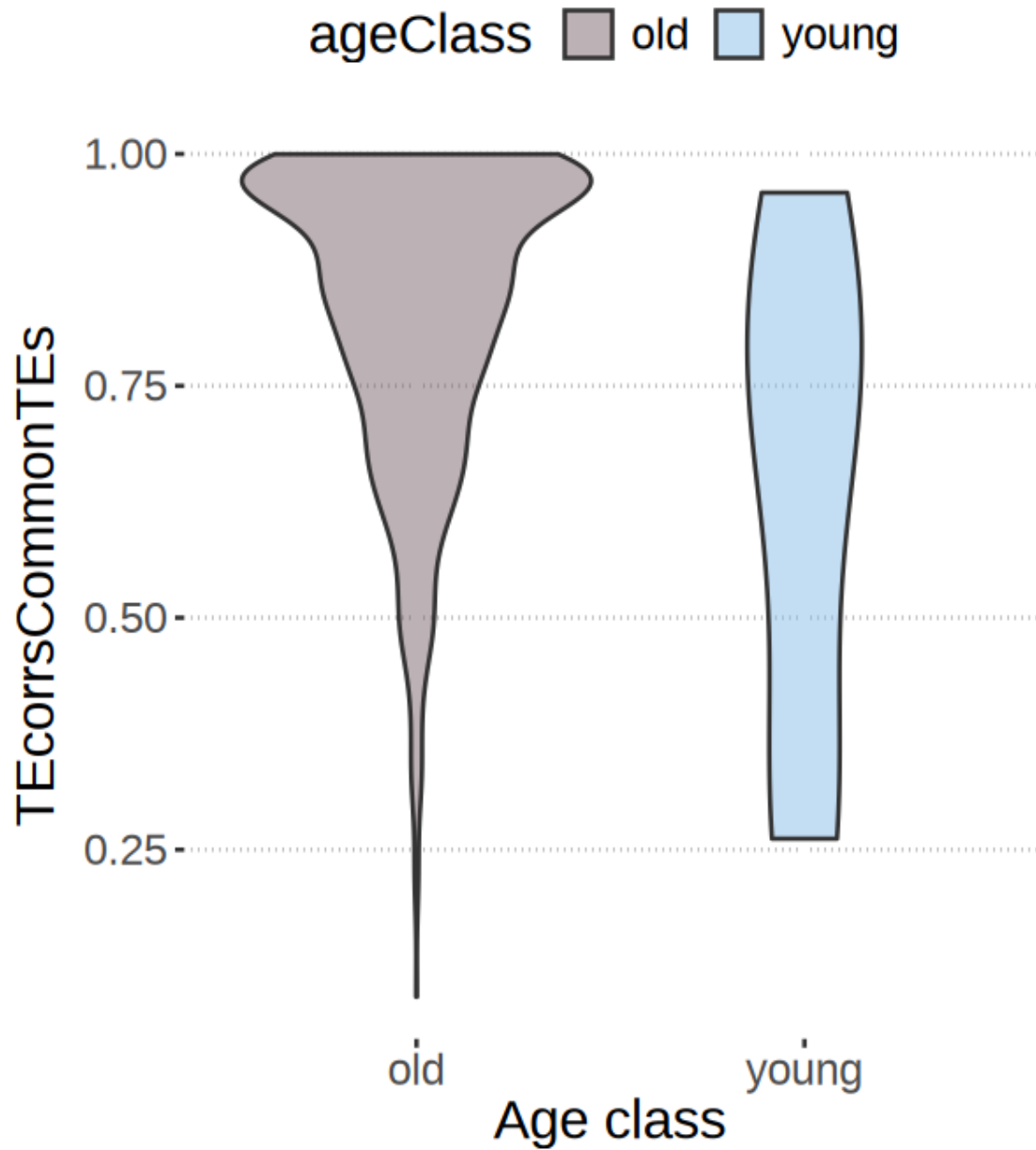


Warning message in `geom_violin(jitter = TRUE, alpha = 0.5)`:
"Ignoring unknown parameters: `jitter`"

Stellarscope_SoloTE

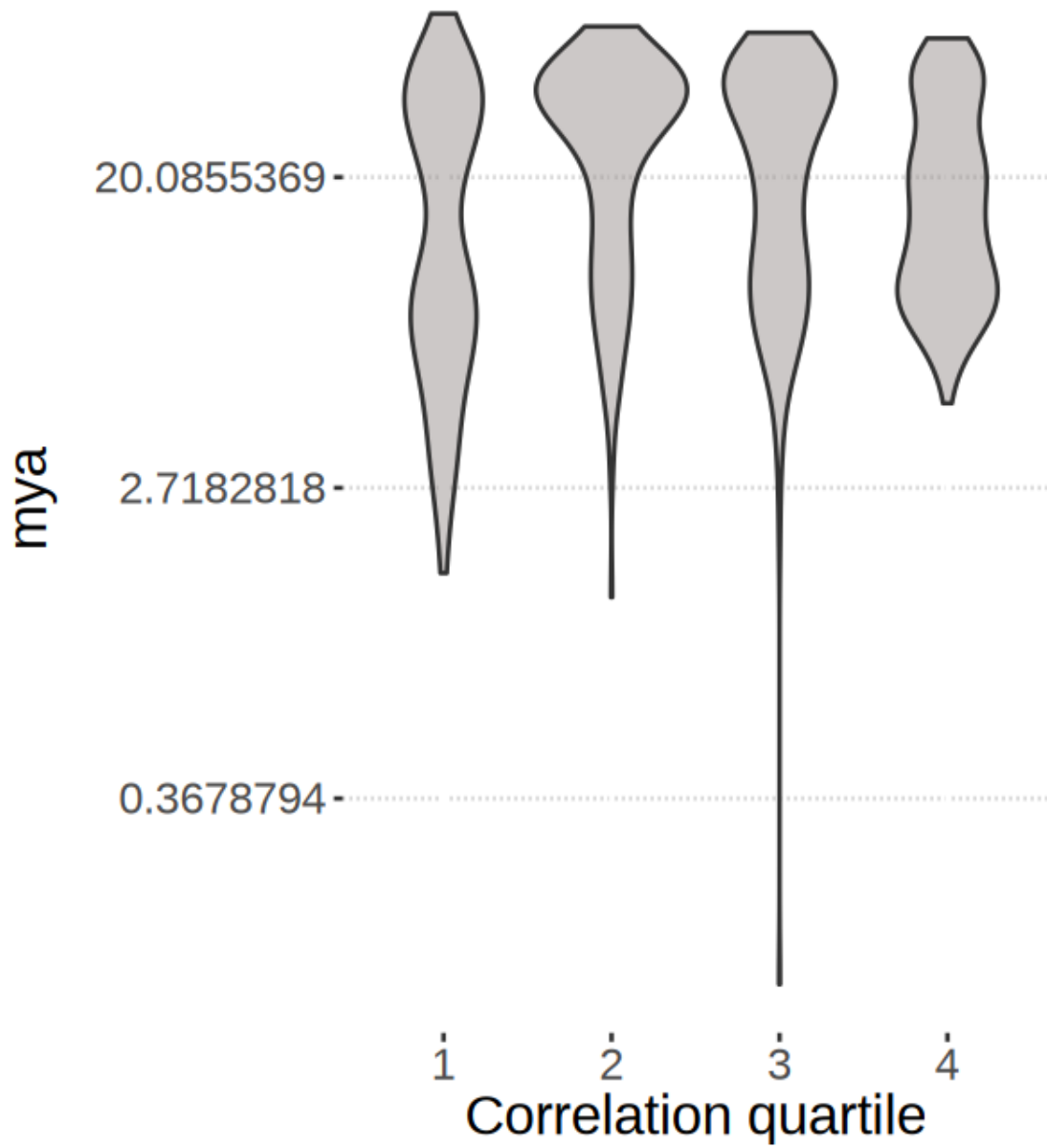


Stellarscope_STARsolo

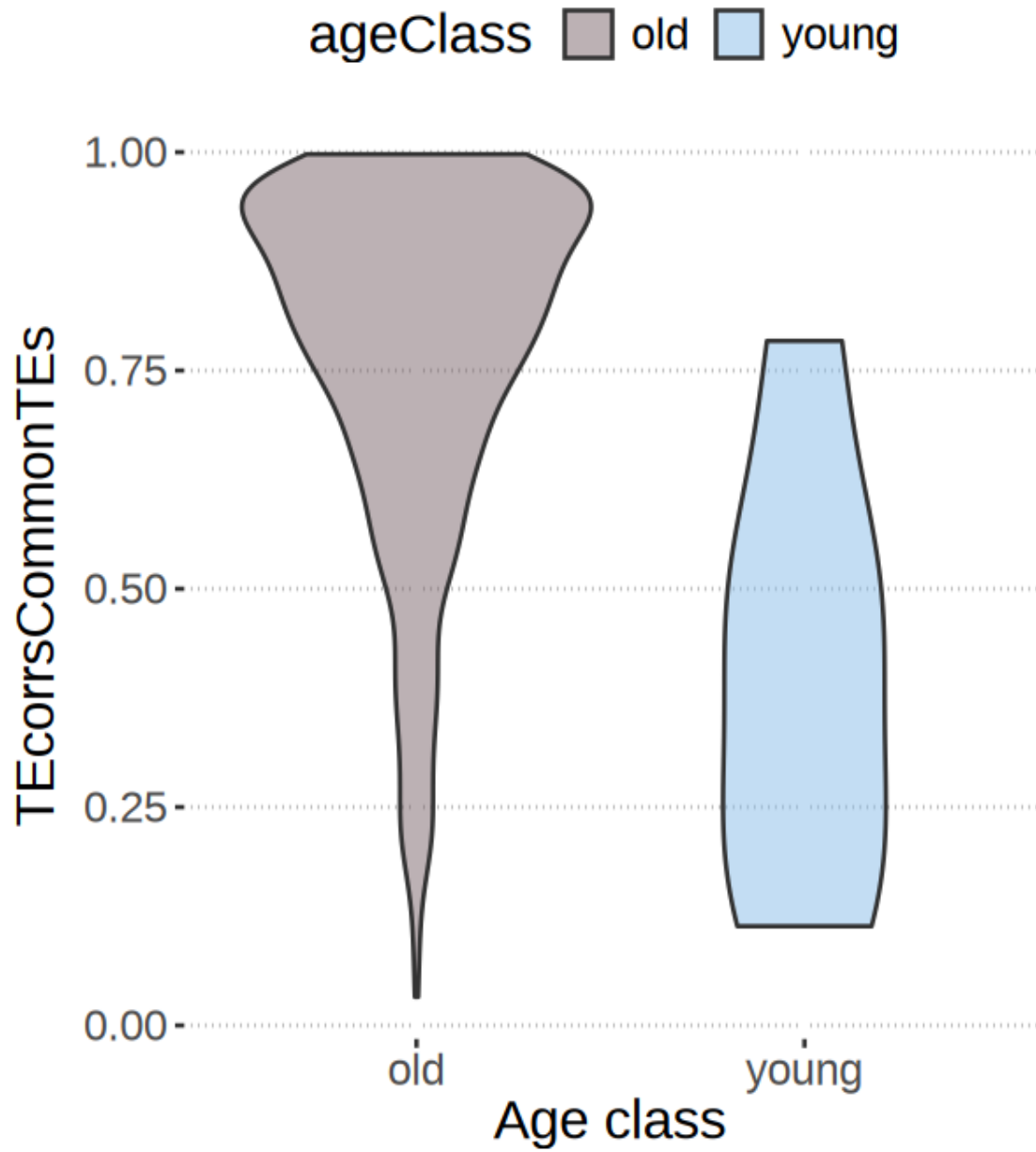


Warning message in `geom_violin(jitter = TRUE, alpha = 0.5)`:
"Ignoring unknown parameters: `jitter`"

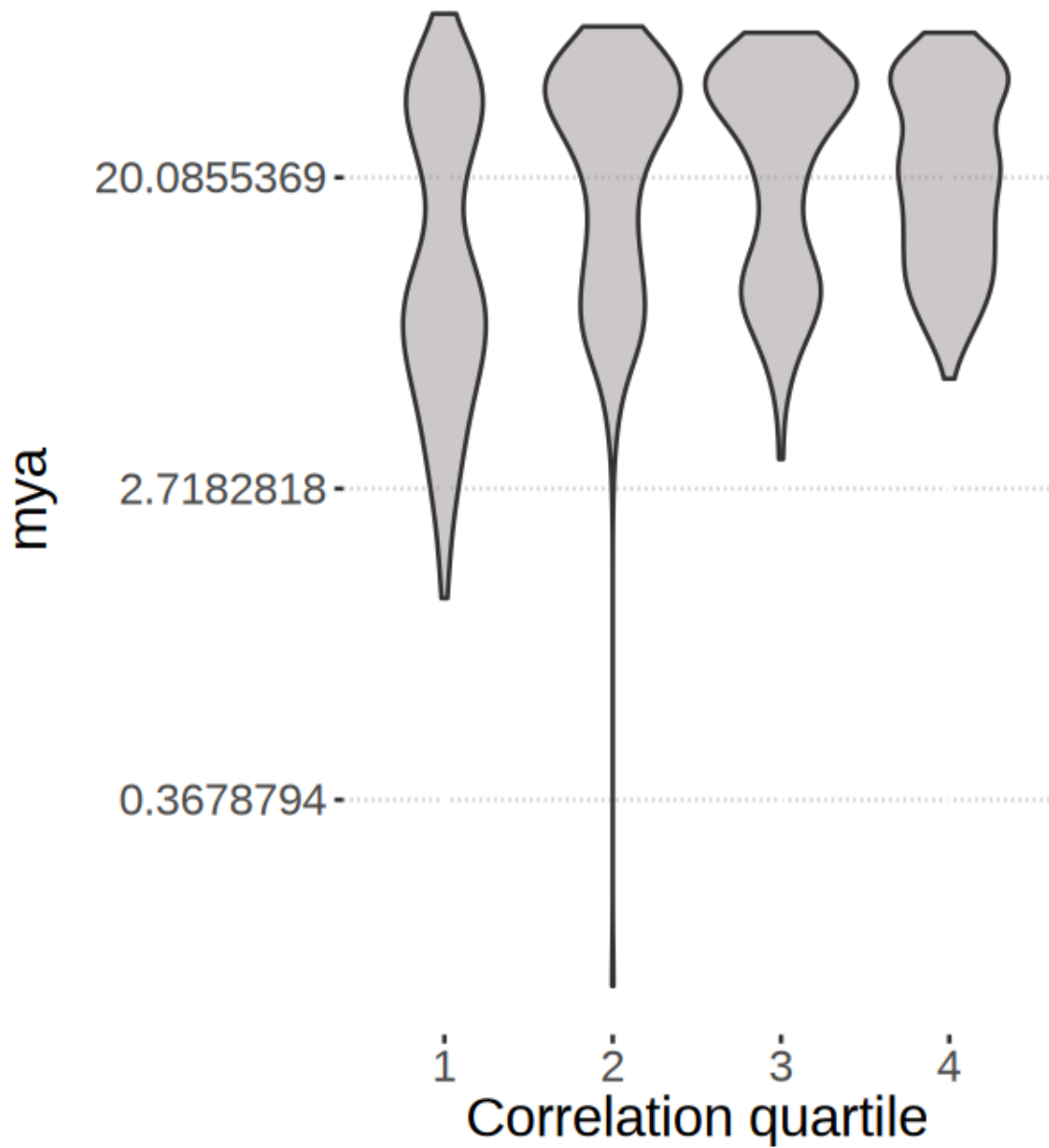
Stellarscope_STARsolo



STARsolo_SoloTE



STARsolo_SoloTE

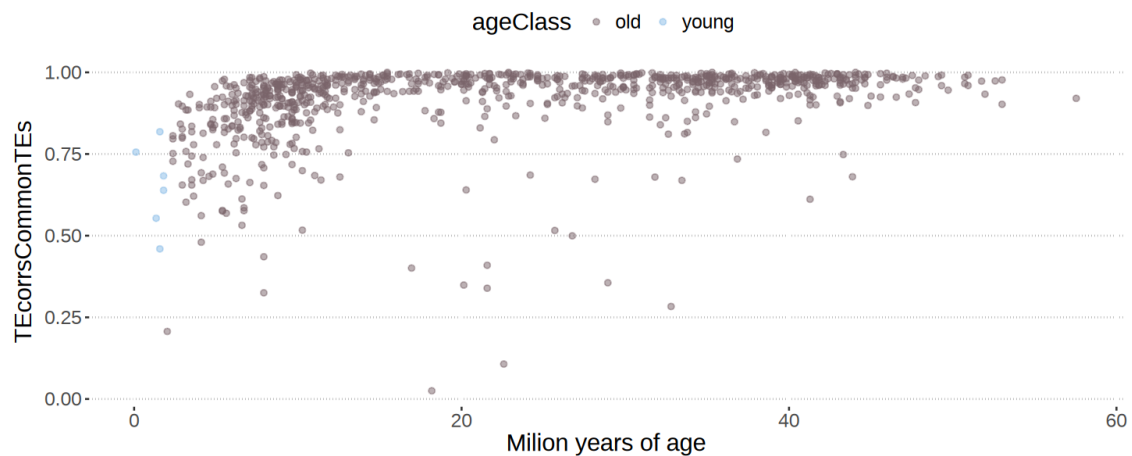
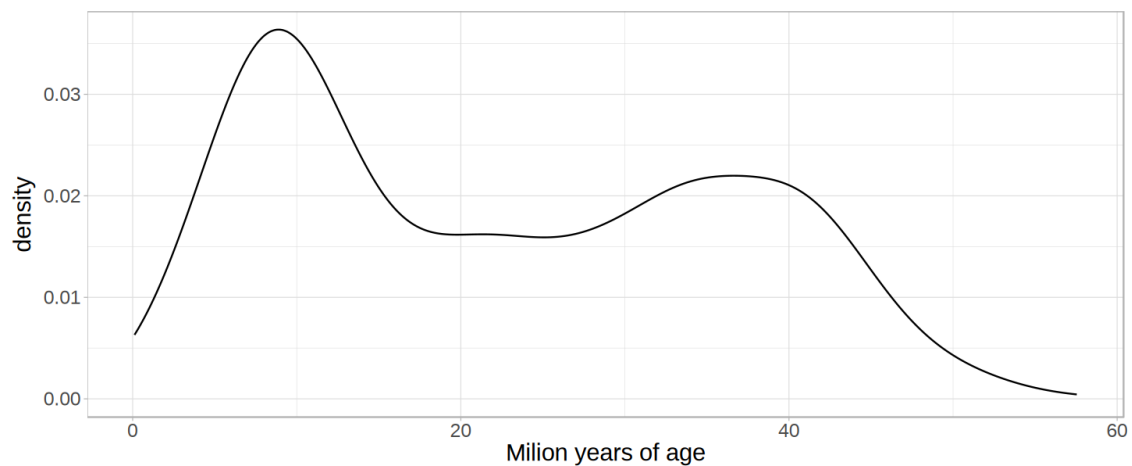


```
[45]: options(repr.plot.width=12, repr.plot.height=5)

ggplot(dfsCorrelationAge[["Stellarscope_SoloTE"]], aes(x=mya)) +
  geom_density(alpha = 0.5) +
  theme_light() +
  theme(text=element_text(size=18)) +
  xlab("Milion years of age")
```

```
ggplot(dfsCorrelationAge[["Stellarscope_SoloTE"]], aes(x=mya, y=TEcorrsCommonTEs, color=ageClass)) +
  geom_point(alpha = 0.5, ) +
  scale_color_manual(values=colorAge) +
  theme_pubclean() +
  theme(text=element_text(size=18)) +
  xlab("Milion years of age")

# this is only for the commonly detected TEs
# the pattern with tes common to SoloTE and Stellarscope was a bit different
```



```
[46]: options(repr.plot.width=7, repr.plot.height=5)

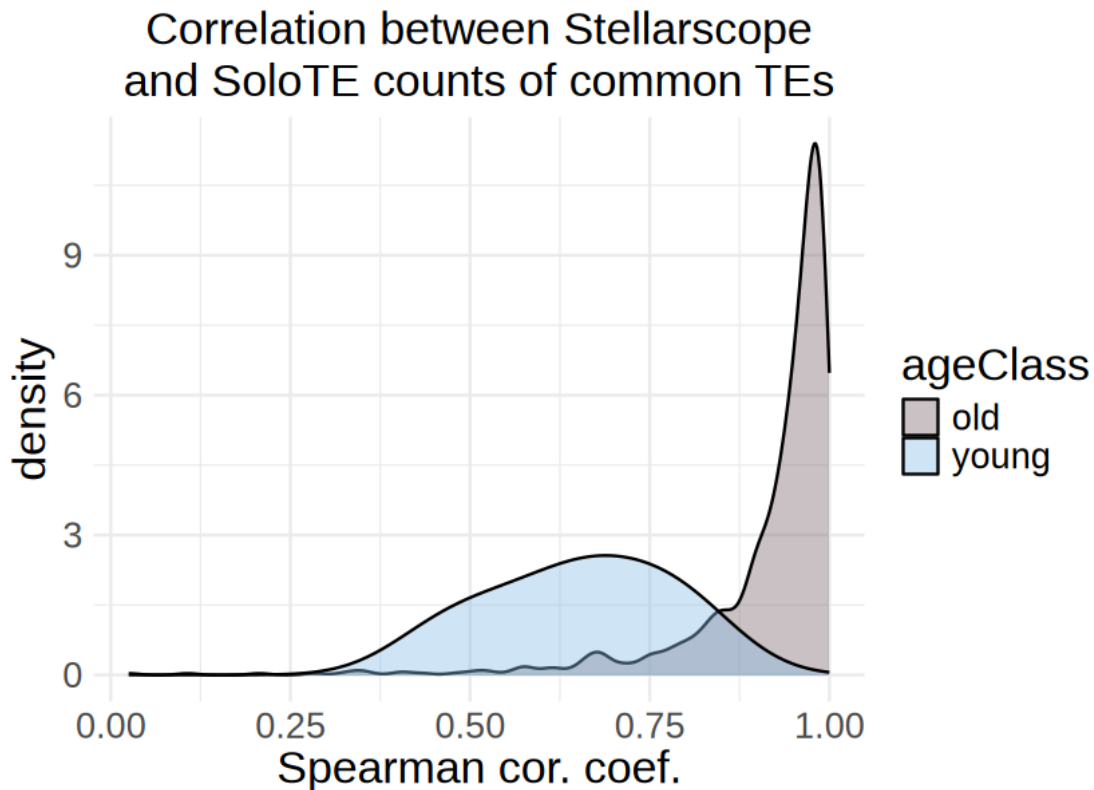
for(combo in names(dfsCorrelationAge)){
```

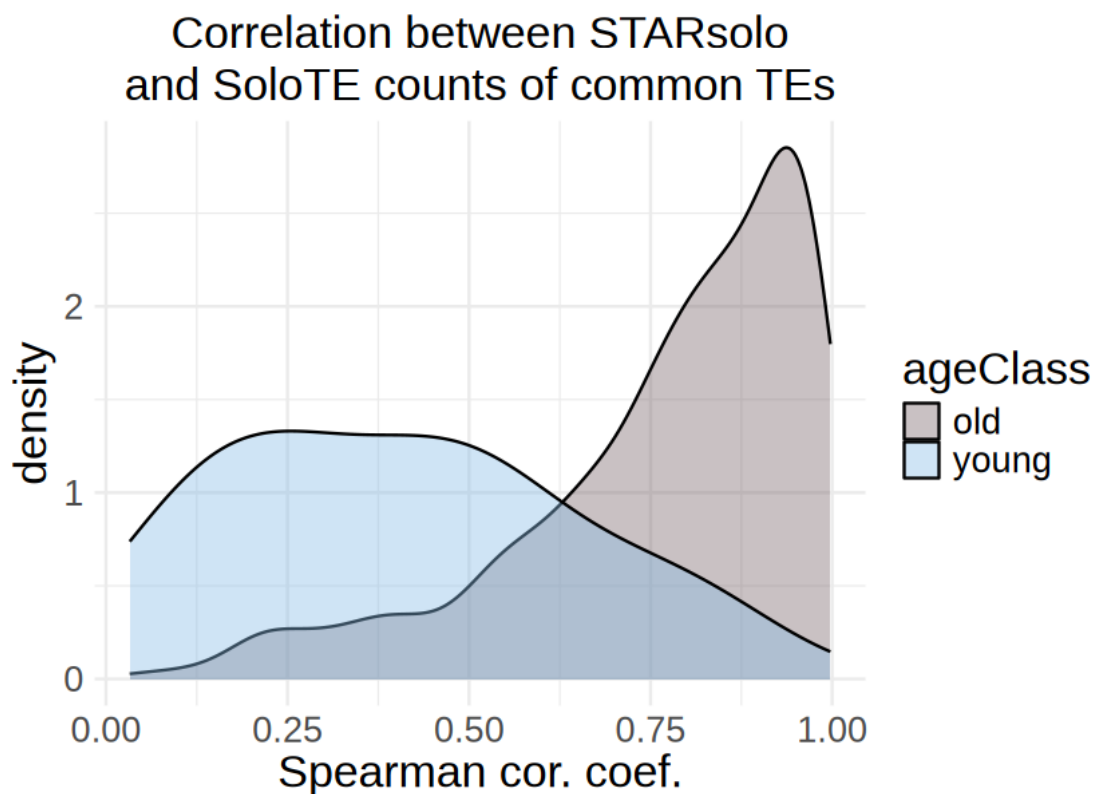
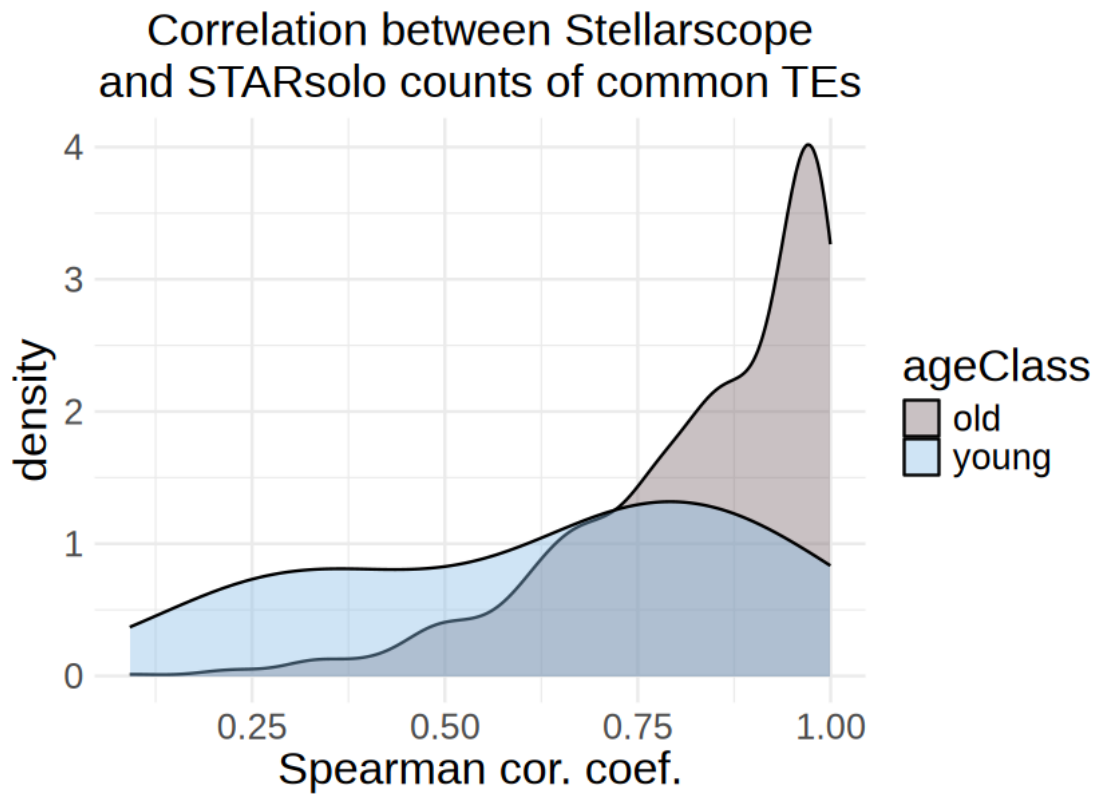
```

tools <- unlist(strsplit(combo,"_"))

show(ggplot(dfsCorrelationAge[[combo]]) + aes(x=TEcorrsCommonTEs,
↪fill=ageClass) +
      geom_density( linewidth = 0.5, alpha = 0.4 ) +
      ggtitle(paste0("Correlation between ", tools[1]),
               subtitle = paste0("and ",tools[2]," counts of common_
↪TEs")) +
      xlab("Spearman cor. coef.") +
      scale_fill_manual(values=colorAge) +
      theme_minimal() +
      theme(text=element_text(size=20),
            plot.title = element_text(size=20, hjust=0.5),
            plot.subtitle = element_text(size=20, hjust=0.5))
    )
ggsave(paste0("figures_",dataset_id,"/correlation_byAge_",combo,".
↪pdf"), width=7, height=5)
}

```





5 Statistics on n features

```
[48]: options(repr.plot.width=3, repr.plot.height=4)

GeomSplitViolin <- ggproto("GeomSplitViolin", GeomViolin,
                           draw_group = function(self, data, ...,
↪draw_quantiles = NULL) {
  data <- transform(data, xminv = x - violinwidth * (x - xmin), xmaxv = x +
↪violinwidth * (xmax - x))
  grp <- data[1, "group"]
  newdata <- plyr::arrange(transform(data, x = if (grp %% 2 == 1) xminv else
↪xmaxv), if (grp %% 2 == 1) y else -y)
  newdata <- rbind(newdata[1, ], newdata, newdata[nrow(newdata), ], newdata[1,
↪])
  newdata[c(1, nrow(newdata) - 1, nrow(newdata)), "x"] <- round(newdata[1, "x"])

  if (length(draw_quantiles) > 0 & !scales::zero_range(range(data$y))) {
    stopifnot(all(draw_quantiles >= 0), all(draw_quantiles <=
      1))
    quantiles <- ggplot2::create_quantile_segment_frame(data, draw_quantiles)
    aesthetics <- data[rep(1, nrow(quantiles)), setdiff(names(data), c("x",
↪"y")), drop = FALSE]
    aesthetics$alpha <- rep(1, nrow(quantiles))
    both <- cbind(quantiles, aesthetics)
    quantile_grob <- GeomPath$draw_panel(both, ...)
    ggplot2::ggname("geom_split_violin", grid::
↪grobTree(GeomPolygon$draw_panel(newdata, ...), quantile_grob))
  }
  else {
    ggplot2::ggname("geom_split_violin", GeomPolygon$draw_panel(newdata, ...))
  }
})

geom_split_violin <- function(mapping = NULL, data = NULL, stat = "ydensity",
↪position = "identity", ...,
                           draw_quantiles = NULL, trim = TRUE, scale =
↪"area", na.rm = FALSE,
                           show.legend = NA, inherit.aes = TRUE) {
  layer(data = data, mapping = mapping, stat = stat, geom = GeomSplitViolin,
        position = position, show.legend = show.legend, inherit.aes = inherit.
↪aes,
        params = list(trim = trim, scale = scale, draw_quantiles =
↪draw_quantiles, na.rm = na.rm, ...))
}
```

```
}
```

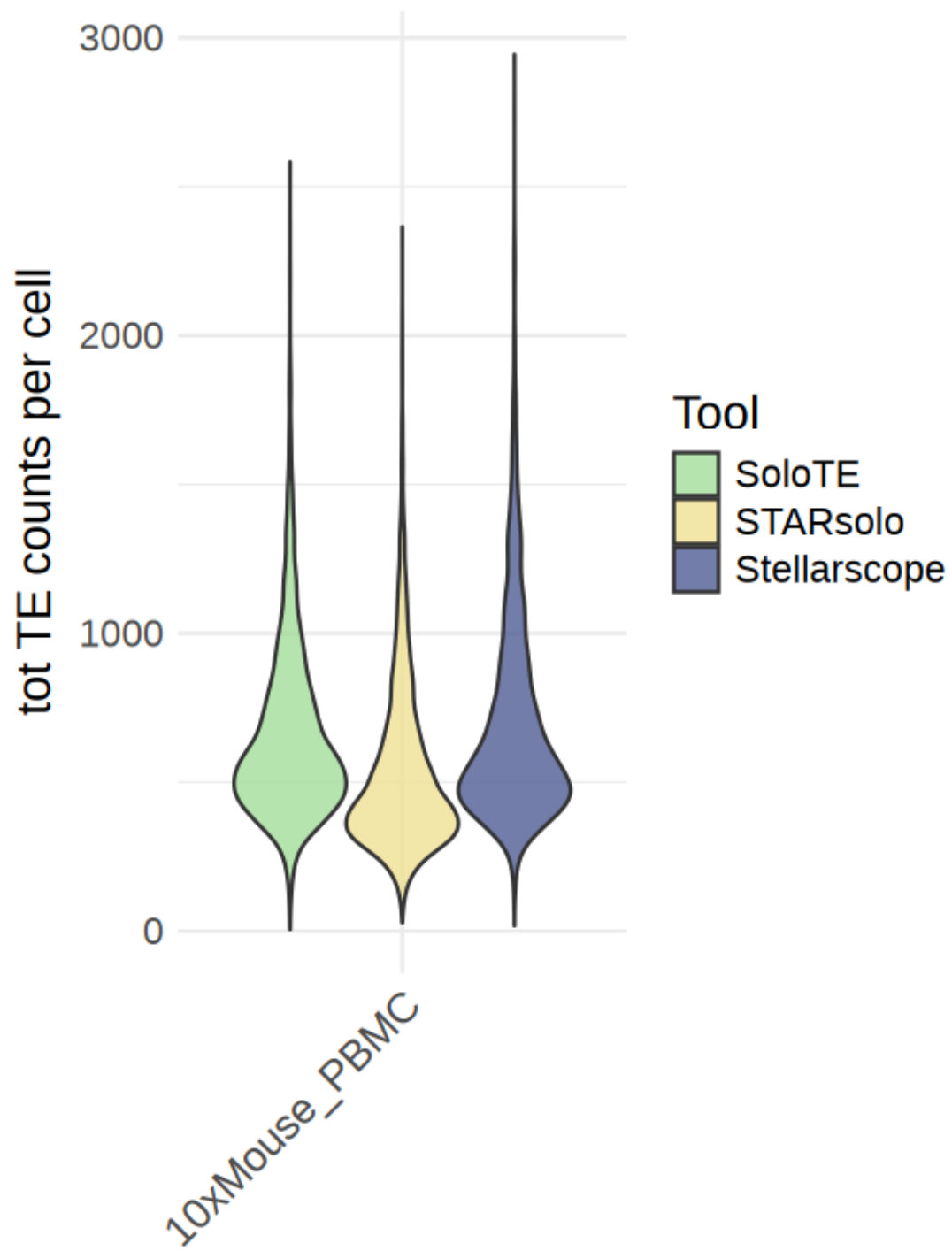
```
[49]: options(repr.plot.width=5, repr.plot.height=7)

nCountsSoloTE <- objTE_soloTE[,commonCellsAll]$nCount_RNA
nCountsStellarscope <- objTE_stellarscope[,commonCellsAll]$nCount_TE
nCountsSTARsolo <- objTE_STARsolo[,commonCellsAll]$nCount_TE

df <- as.data.frame(rbind(cbind(nCountsSoloTE,"SoloTE",␣
  ↪apply(strsplit(names(nCountsSoloTE), "_"),'[,1)),
                                cbind(nCountsStellarscope,"Stellarscope",␣
  ↪apply(strsplit(names(nCountsStellarscope), "_"),'[,1)),
                                cbind(nCountsSTARsolo,"STARsolo",␣
  ↪apply(strsplit(names(nCountsSTARsolo), "_"),'[,1)))))

colnames(df) <- c("nCounts_TEs","Tool","CB")
df$Sample <- dataset_id
df$nCounts_TEs <- as.numeric(df$nCounts_TEs)

ggplot(df, aes(x=Sample, fill=Tool, y=nCounts_TEs)) +
  geom_violin(scale = "width", alpha = 0.8, color="grey20") +
  scale_fill_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("tot TE counts per cell") +
  theme_minimal() +
  theme(text=element_text(size=17),
        plot.title = element_text(size=17, hjust=0.5),
        plot.subtitle = element_text(size=17, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/totCounts_SoloTEvsStellarscopevsSTARsolo.
  ↪pdf"), device = "pdf", width=5, height=7)
```



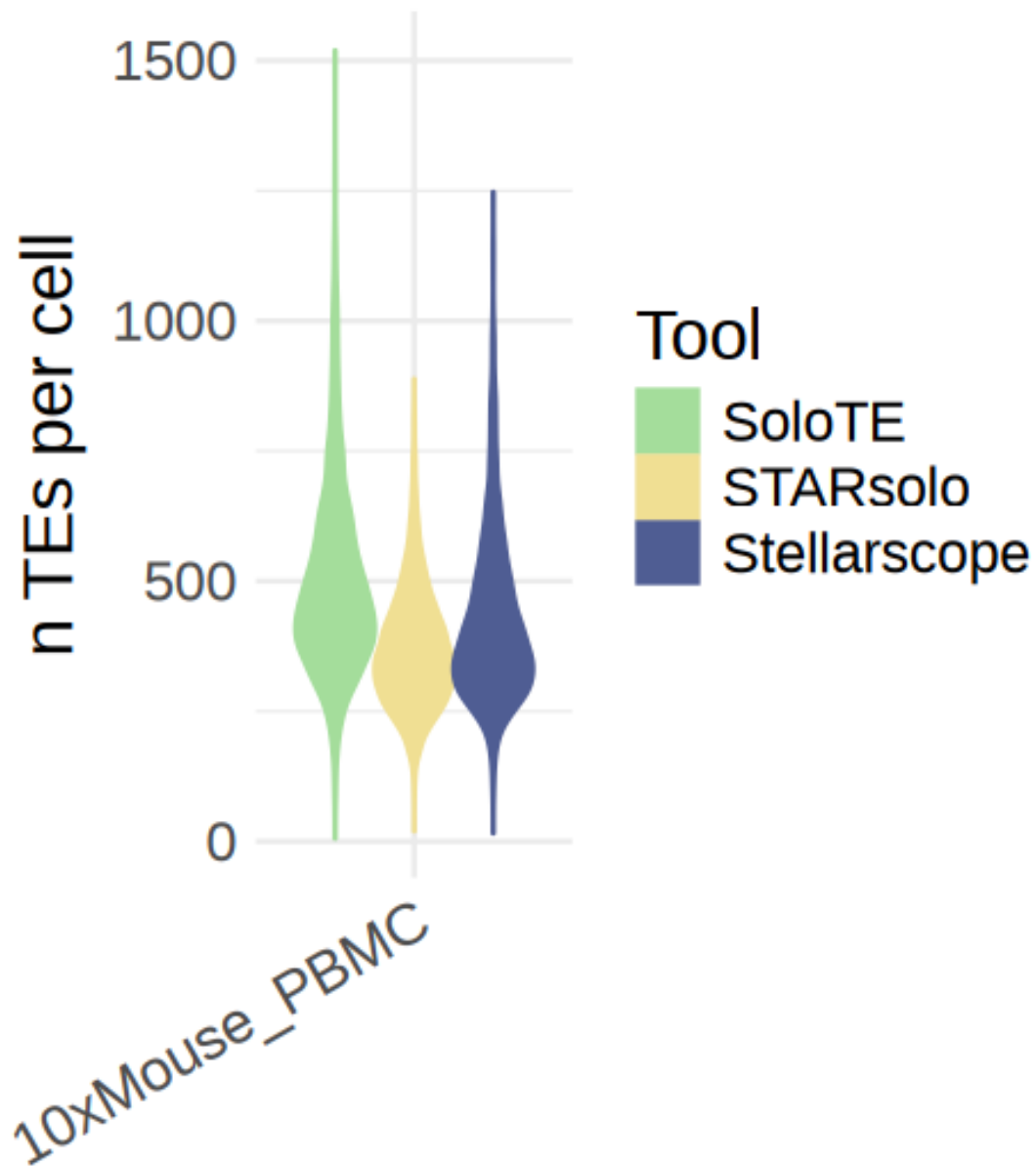

```
[50]: options(repr.plot.width=4, repr.plot.height=5)

nFeaturesSoloTE <- objTE_soloTE[,commonCellsAll]$nFeature_RNA
nFeaturesStellarscope <- objTE_stellarscope[,commonCellsAll]$nFeature_TE
nFeaturesSTARsolo <- objTE_STARsolo[,commonCellsAll]$nFeature_TE

df <- as.data.frame(rbind(cbind(nFeaturesSoloTE,"SoloTE",
  ↪apply(strsplit(names(nFeaturesSoloTE), "_"),'[,1)),
                        cbind(nFeaturesStellarscope,"Stellarscope",
  ↪apply(strsplit(names(nFeaturesStellarscope), "_"),'[,1)),
                        cbind(nFeaturesSTARsolo,"STARsolo",
  ↪apply(strsplit(names(nFeaturesSTARsolo), "_"),'[,1)))))

colnames(df) <- c("nFeatures_TEs","Tool","Sample")
df$Sample <- dataset_id
df$nFeatures_TEs <- as.numeric(df$nFeatures_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))

ggplot(df, aes(x=Sample, fill=Tool, color=Tool, y=nFeatures_TEs)) +
  geom_violin(scale = "width") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("n TEs per cell") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=30, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
  ↪totFeatures_SoloTEvsStellarscopevsSTARsolo_violin.pdf"), device = "pdf",
  ↪width=4, height=5)
```



```
[51]: objTE_stellarscope
      mat <- GetAssayData(objTE_stellarscope, layer="counts")
      sum(rowSums(mat > 0) >= 1)
```

An object of class Seurat

4123 features across 4787 samples within 1 assay
 Active assay: RNA (4123 features, 4000 variable features)
 3 layers present: counts, data, scale.data
 2 dimensional reductions calculated: pca, umap
 4123

```
[52]: # save lists of TEs expressed in each sample
nTEs_soloTE <- list()
TEs_soloTE <- list()
for(ident in unique(objTE_soloTE$orig.ident)){
  mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident],
    layer="counts")
  nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= 15)
  TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= 15]
}
TEs_soloTE_df <- stack(TEs_soloTE)
colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

nTEs_stellarscope <- list()
TEs_stellarscope <- list()
for(ident in unique(objTE_stellarscope$orig.ident)){
  mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.
    ident==ident], layer="counts")
  nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= 15)
  TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= 15]
}
TEs_stellarscope_df <- stack(TEs_stellarscope)
colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")

nTEs_star <- list()
TEs_star <- list()
for(ident in unique(objTE_STARsolo$orig.ident)){
  mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident],
    layer="counts")
  nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= 15)
  TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= 15]
}
TEs_star_df <- stack(TEs_stellarscope)
colnames(TEs_star_df) <- c("locus", "orig.ident")

options(repr.plot.width=4, repr.plot.height=5)
# N TEs per tool
df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE),
  rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
  cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope),
  rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
```

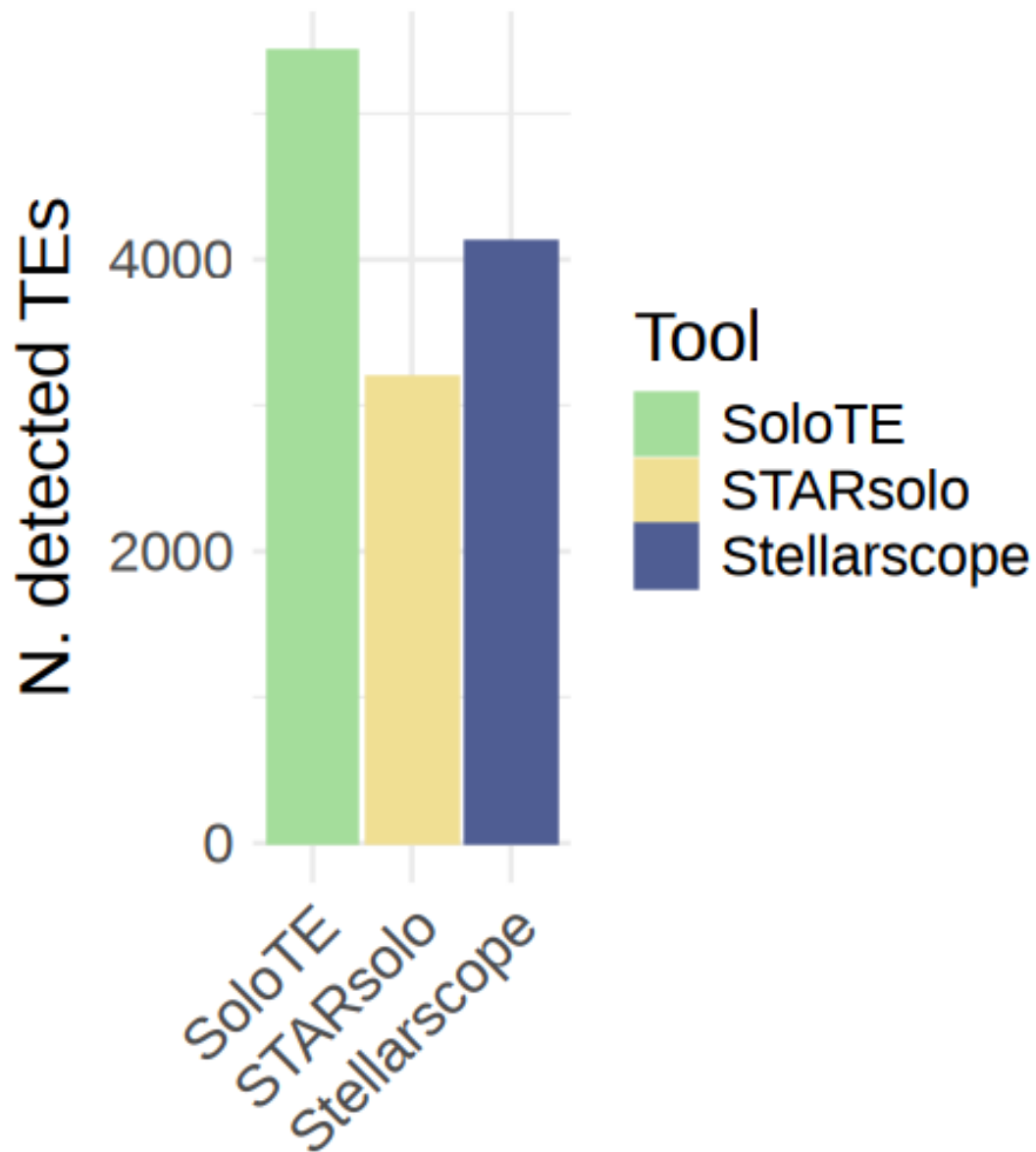
```

        cbind(unlist(nTEs_star), names(nTEs_star),
↪rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
)

colnames(df) <- c("n_TEs", "Sample", "Tool")
df <- as.data.frame(df)
df$n_TEs <- as.numeric(df$n_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))
df
ggplot(df, aes(x=Tool,fill=Tool, color=Tool, y=n_TEs)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle("") +
  xlab("") +
  ylab("N. detected TEs") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
↪nTEs_SoloTEvsStellarscopevsSTARsolo_barplot.pdf"), device = "pdf", width =
↪4, height=5)

```

		n_TEs	Sample	Tool
		<dbl>	<fct>	<chr>
A data.frame: 3 × 3	PBMC	5430	PBMC	SoloTE
	PBMC.1	4123	PBMC	Stellarscope
	STARsolo_TE_EM	3182	STARsolo_TE_EM	STARsolo



```
[53]: library(UpSetR)

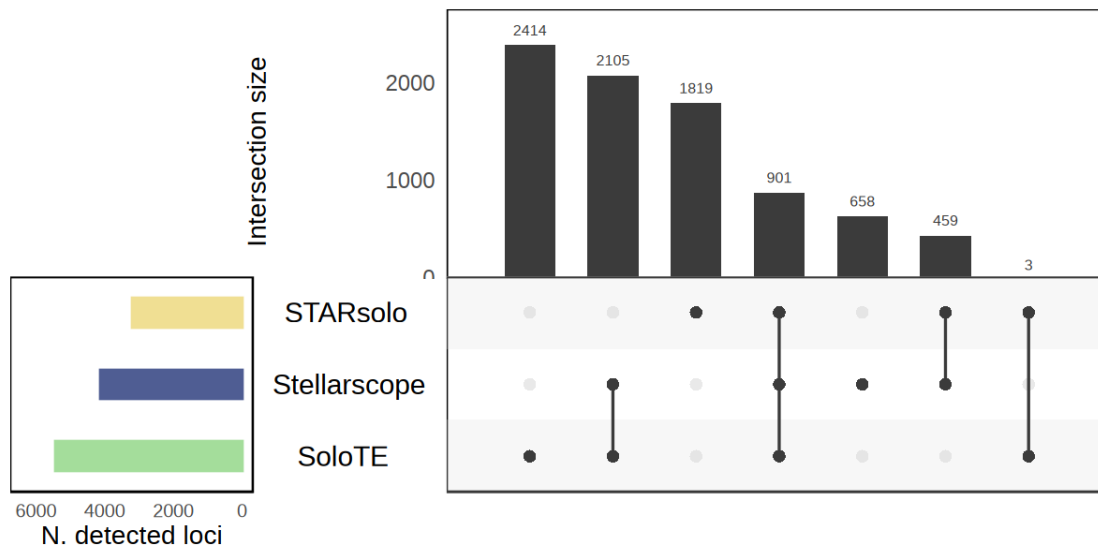
options(repr.plot.width=10, repr.plot.height=5)
```

```

detectedList <- list(SoloTE = tes_soloTE_conv, Stellarscope = tes_stellarscope_conv,
  ↪ tes_stellarscope_conv, STARsolo = tes_starsolo_conv)

UpSetR::upset(fromList(detectedList), nintersects = 30,
  sets=names(colorTools), keep.order = T, sets.bar.color=colorTools,
  text.scale = c(2, 2, 2, 1.65, 2.5, 1.5),
  order.by=c("freq"),
  point.size=3, mb.ratio = c(0.5, 0.5),
  set_size.show=F, set_size.scale_max=max(lengths(detectedList))+1000, #show.numbers = F,
  ↪ nsets=length(objList)+1,
  sets.x.label="N. detected loci",
  mainbar.y.label="Intersection size")
  ggsave(paste0("figures_",dataset_id,"/
  ↪ nTEs_SoloTEvsStellarscopevsSTARsolo_barplot.pdf"), device = "pdf", width = 10, height=5)

```



```

[54]: options(repr.plot.width=5, repr.plot.height=5)

library(eulerr)

# remove NAs
sum(is.na(detectedList[["SoloTE"]])) # ?
detectedList[["SoloTE"]][is.na(detectedList[["SoloTE"]])] <- (1:sum(is.
  ↪ na(detectedList[["SoloTE"]]))
sum(is.na(detectedList[["SoloTE"]]))

# Convert lists to sets and compute intersections

```

```

fit <- euler(detectedList)

# Plot with proportional areas
plot(fit,
      fills = colorTools,
      alpha = 0.5,
      labels = list(font = 2, cex = 1.5),      # <-- increase set label size
      quantities = list(cex = 1.4),           # <-- increase count size
      main.cex = list(cex=2),
      edges = list(col = "white", lwd = 2),
      # labels = TRUE,
      # quantities = TRUE,
      main = "Overlap across detected TEs")

pdf(paste0("figures_", dataset_id, "/overlap_euler_withSTAR.pdf"), width=5, h
    height=5)
plot(fit,
      fills = colorTools,
      alpha = 0.5,
      labels = list(font = 2, cex = 1.5),      # <-- increase set label size
      quantities = list(cex = 1.4),           # <-- increase count size
      main.cex = list(cex=2),
      edges = list(col = "white", lwd = 2),
      # labels = TRUE,
      # quantities = TRUE,
      main = "Overlap across detected TEs")
dev.off()

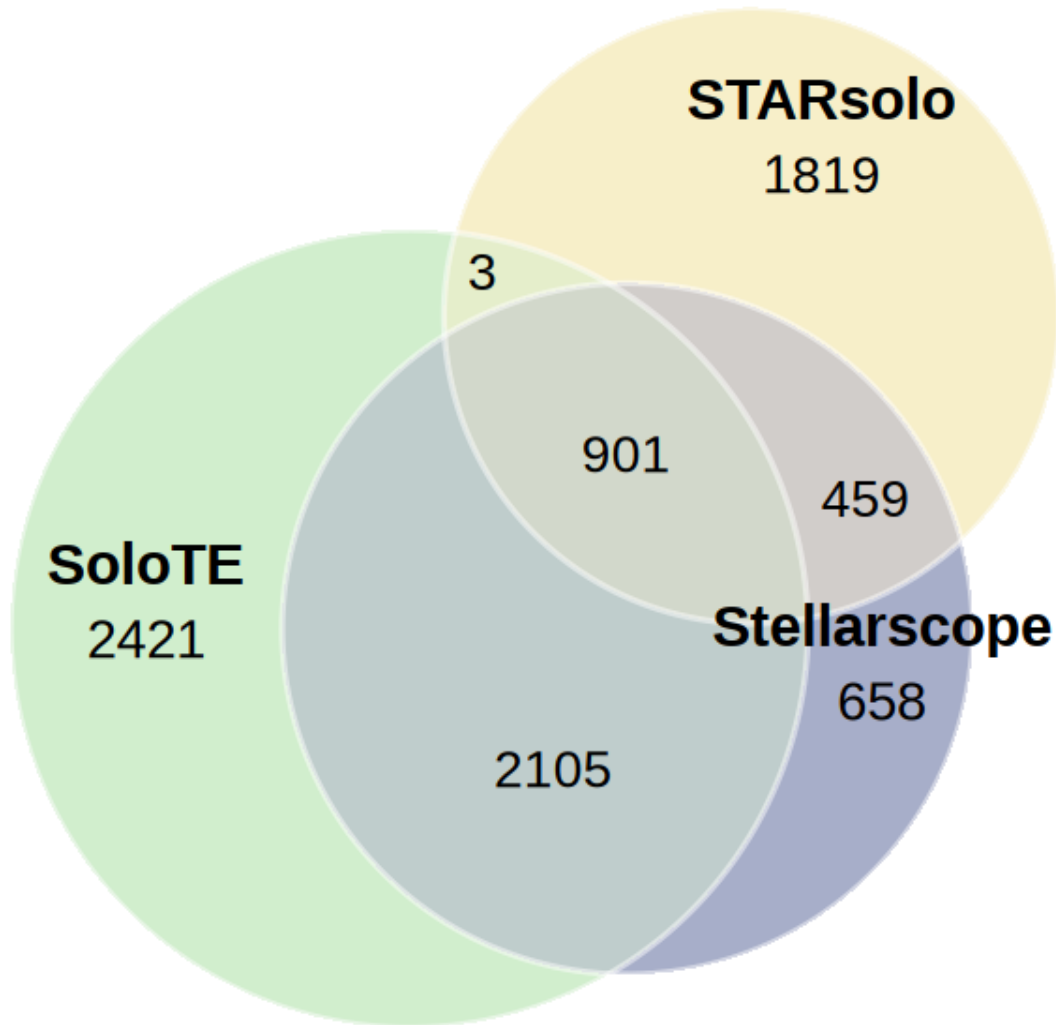
```

8

0

agg__record__1909893419: 2

Overlap across detected TEs



```
[55]: # Number of TEs detected above a certain threshold of cells ( 5% of the dataset)

cellNum <- round(length(Cells(objTE_soloTE))*0.05)
# save lists of TEs expressed in each sample
nTEs_soloTE <- list()
TEs_soloTE <- list()

for(ident in unique(objTE_soloTE$orig.ident)){
  mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident],
    layer="counts")
  nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
```



```

TEs_soloTE_df <- stack(TEs_soloTE)
colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

nTEs_stellarscope <- list()
TEs_stellarscope <- list()
for(ident in unique(objTE_stellarscope$orig.ident)){
  mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.
    ↪ident==ident], layer="counts")
  nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_stellarscope_df <- stack(TEs_stellarscope)
colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")

nTEs_star <- list()
TEs_star <- list()
for(ident in unique(objTE_STARsolo$orig.ident)){
  mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident],
    ↪layer="counts")
  nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_star_df <- stack(TEs_stellarscope)
colnames(TEs_star_df) <- c("locus", "orig.ident")

options(repr.plot.width=4, repr.plot.height=5)
# N TEs per tool
df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE),
    ↪rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
             cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope),
    ↪rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
             cbind(unlist(nTEs_star), names(nTEs_star),
    ↪rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
)

colnames(df) <- c("n_TEs", "Sample", "Tool")
df <- as.data.frame(df)
df$n_TEs <- as.numeric(df$n_TEs)
df$Sample <- factor(df$Sample, levels=unique(df$Sample))
df
ggplot(df, aes(x=Tool,fill=Tool, color=Tool, y=n_TEs)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle(paste0("Loci detected in 5% of cells")) +
  xlab("") +
  ylab("N. detected TEs") +

```

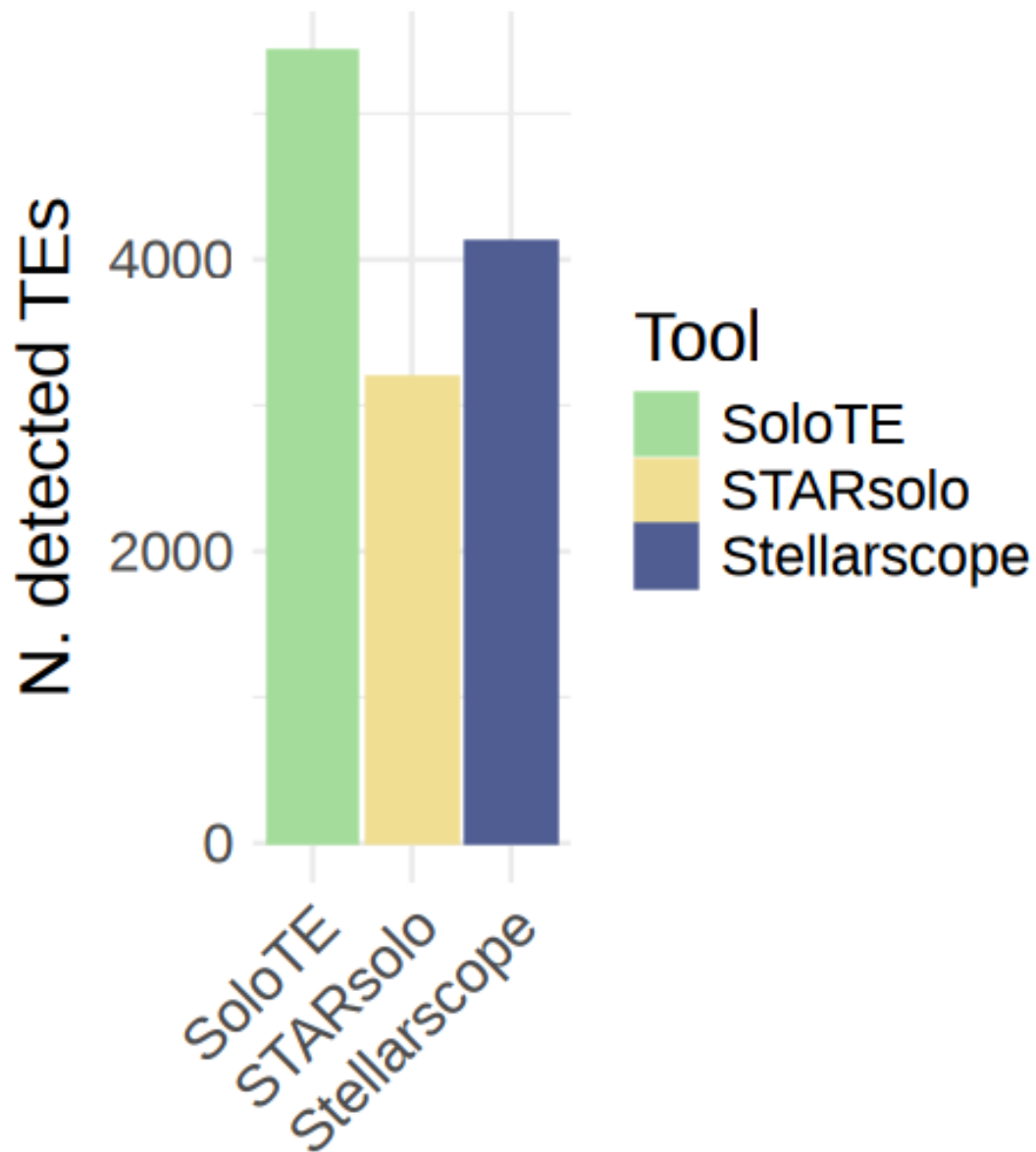
```

theme_minimal() +
theme(text=element_text(size=18),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
↳nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_thr5percCells.pdf"), device =
↳"pdf", width = 5, height=5)

```

		n_TEs <dbl>	Sample <fct>	Tool <chr>
A data.frame: 3 × 3	PBMC	5430	PBMC	SoloTE
	PBMC.1	4123	PBMC	Stellarscope
	STARsolo_TE_EM	3182	STARsolo_TE_EM	STARsolo

Loci detected in 5% of cells



```
[56]: # Number of TEs detected above a certain threshold of cells ( 5% of the dataset)

cellNum <- round(length(Cells(objTE_soloTE))*0.02)
# save lists of TEs expressed in each sample
nTEs_soloTE <- list()
```

```

TEs_soloTE <- list()

for(ident in unique(objTE_soloTE$orig.ident)){
  mat <- GetAssayData(objTE_soloTE[,objTE_soloTE$orig.ident==ident],
    layer="counts")
  nTEs_soloTE[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_soloTE[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_soloTE_df <- stack(TEs_soloTE)
colnames(TEs_soloTE_df) <- c("locus", "orig.ident")

nTEs_stellarscope <- list()
TEs_stellarscope <- list()
for(ident in unique(objTE_stellarscope$orig.ident)){
  mat <- GetAssayData(objTE_stellarscope[,objTE_stellarscope$orig.
    ident==ident], layer="counts")
  nTEs_stellarscope[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_stellarscope[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_stellarscope_df <- stack(TEs_stellarscope)
colnames(TEs_stellarscope_df) <- c("locus", "orig.ident")

nTEs_star <- list()
TEs_star <- list()
for(ident in unique(objTE_STARsolo$orig.ident)){
  mat <- GetAssayData(objTE_STARsolo[,objTE_STARsolo$orig.ident==ident],
    layer="counts")
  nTEs_star[[ident]] <- sum(rowSums(mat > 0) >= cellNum)
  TEs_star[[ident]] <- rownames(mat)[rowSums(mat > 0) >= cellNum]
}
TEs_star_df <- stack(TEs_stellarscope)
colnames(TEs_star_df) <- c("locus", "orig.ident")

options(repr.plot.width=4, repr.plot.height=5)
# N TEs per tool
df <- rbind( cbind(unlist(nTEs_soloTE), names(nTEs_soloTE),
  rep("SoloTE",length(unique(objTE_soloTE$orig.ident)))),
  cbind(unlist(nTEs_stellarscope), names(nTEs_stellarscope),
  rep("Stellarscope",length(unique(objTE_stellarscope$orig.ident)))),
  cbind(unlist(nTEs_star), names(nTEs_star),
  rep("STARsolo",length(unique(objTE_STARsolo$orig.ident))))
)

colnames(df) <- c("n_TEs", "Sample", "Tool")
df <- as.data.frame(df)
df$n_TEs <- as.numeric(df$n_TEs)

```

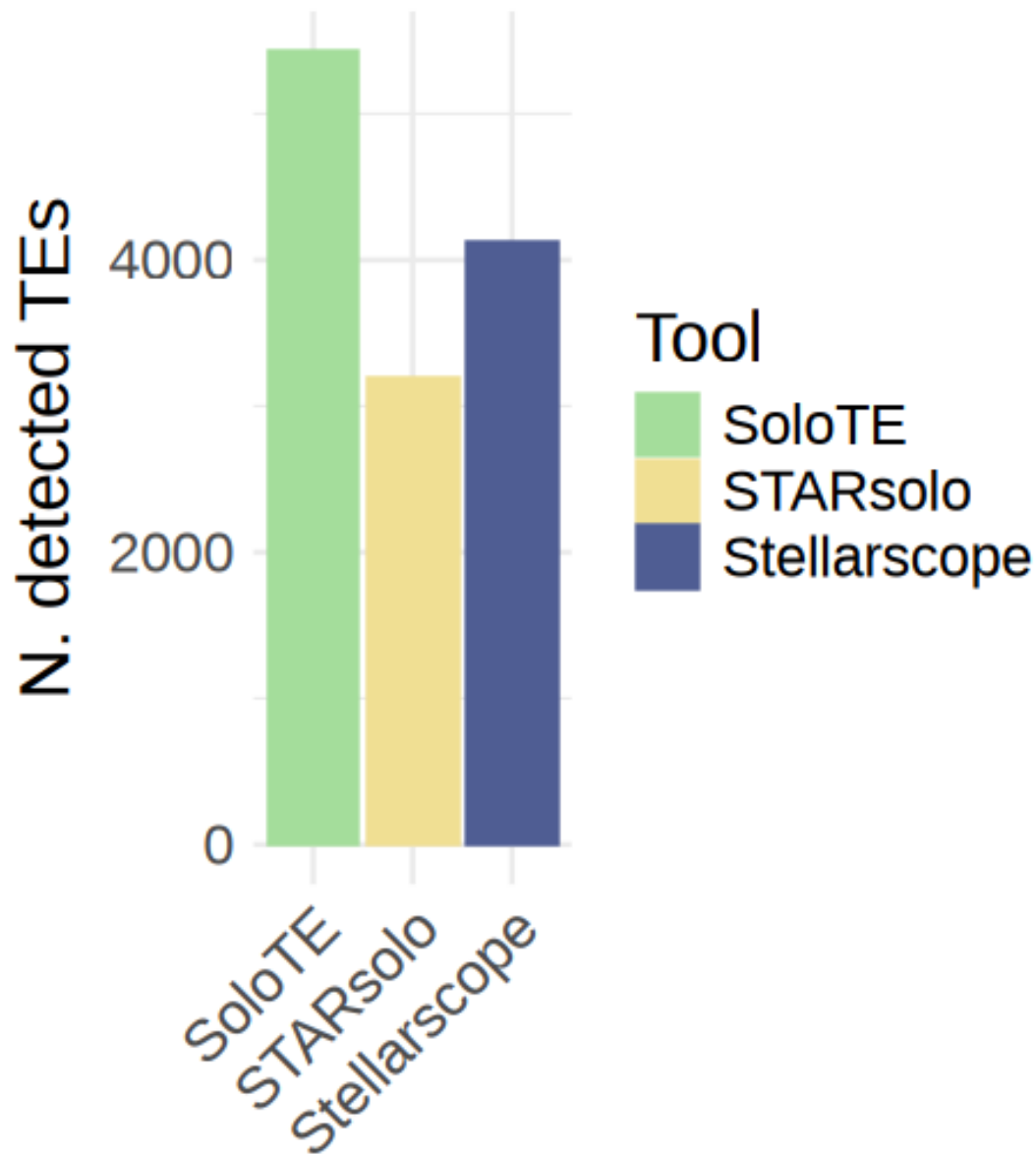
```

df$Sample <- factor(df$Sample, levels=unique(df$Sample))
df
ggplot(df, aes(x=Tool,fill=Tool, color=Tool, y=n_TEs)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values=colorTools)+
  scale_color_manual(values=colorTools)+
  ggtitle(paste0("Loci detected in 2% of cells")) +
  xlab("") +
  ylab("N. detected TEs") +
  theme_minimal() +
  theme(text=element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.x = element_text(size=15, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/
↳nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_thr2percCells.pdf"), device =
↳"pdf", width = 5, height=5)

```

		n_TEs	Sample	Tool
		<dbl>	<fct>	<chr>
A data.frame: 3 × 3	PBMC	5430	PBMC	SoloTE
	PBMC.1	4123	PBMC	Stellarscope
	STARsolo_TE_EM	3182	STARsolo_TE_EM	STARsolo

Loci detected in 2% of cells



```
[57]: options(repr.plot.width=5, repr.plot.height=7)
TEs_soloTE_df$age <- conversionTable[match(TEs_soloTE_df$locus,
  ↪conversionTable$soloteID), "ageClass"]
TEs_stellarscope_df$age <- conversionTable[match(TEs_stellarscope_df$locus,
  ↪conversionTable$stellarscopeID), "ageClass"]
```

```

TEs_star_df$age <- conversionTable[match(TEs_star_df$locus,
↳conversionTable$stellarscopeID), "ageClass"]

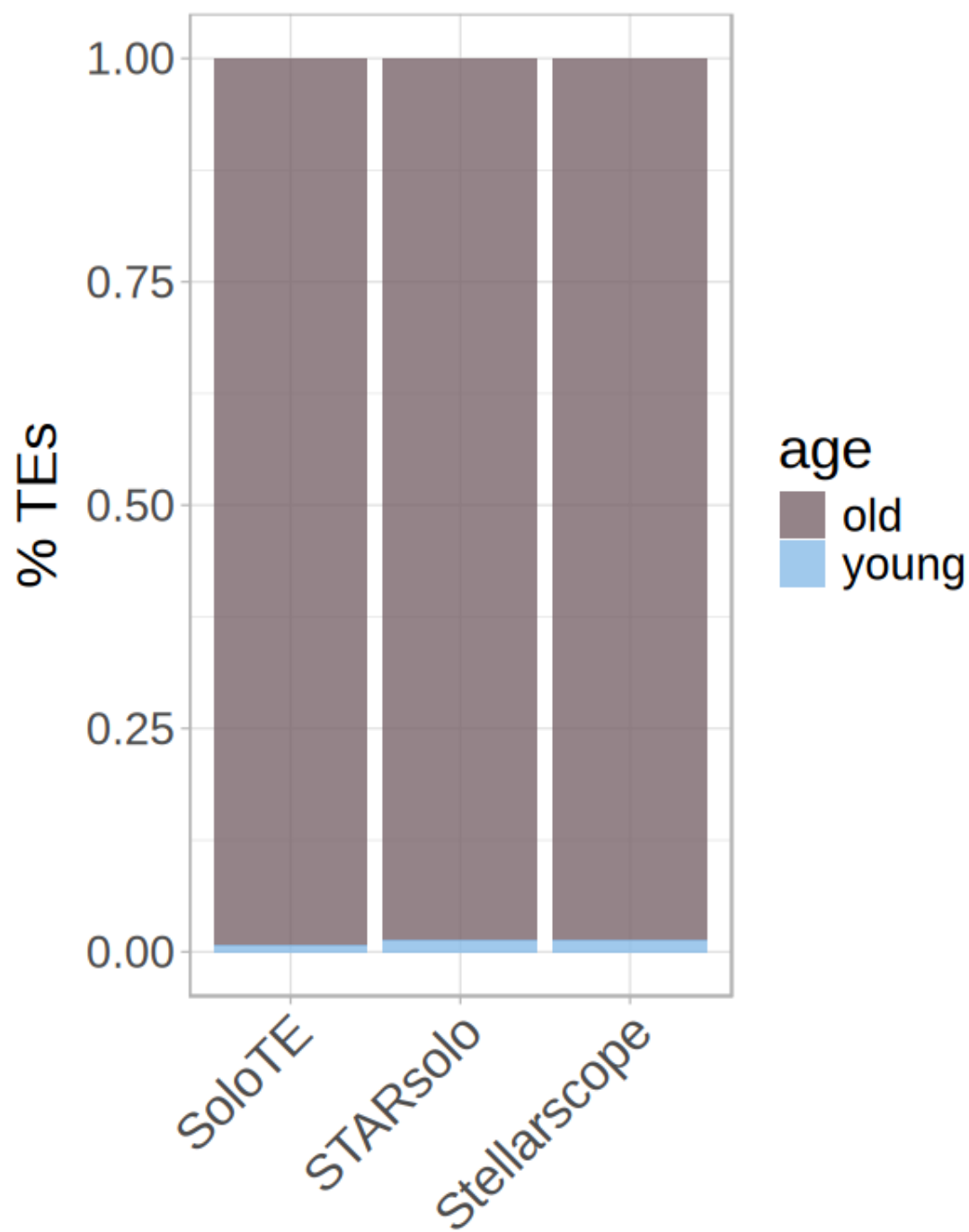
TEs_soloTE_df$tool <- "SoloTE"
TEs_stellarscope_df$tool <- "Stellarscope"
TEs_star_df$tool <- "STARsolo"
df <- rbind(TEs_soloTE_df, TEs_stellarscope_df, TEs_star_df)

df <- df[!is.na(df$age),] # why are there NAs? Didn't I remove TEs not in the
↳annotation beforehand?

ggplot(df, aes(x=tool, fill=age))+
  geom_bar(position="fill", alpha = 0.8) +
  scale_fill_manual(values=colorAge) +
  ggtitle("") +
  xlab("") +
  ylab("% TEs") +
  theme_light() +
  theme(text=element_text(size=20),
        strip.text.x = element_text(size=22, hjust=0.5),
        strip.background = element_rect(fill="#7E7E7E"),
        axis.text.x = element_text(size=18, angle=45, hjust=1))

ggsave(paste0("figures_", dataset_id, "/
↳nTEs_SoloTEvsStellarscopevsSTARsolo_barplot_age_tool_celltype.pdf"), device=
↳ "pdf", width=5, height=7)

```




```

[58]: options(repr.plot.width=8, repr.plot.height=7)
soloTEageFamily <- table(conversionTable[match(Features(objTE_soloTE),
  ↪conversionTable$soloteID), "ageClass"],
  ↪conversionTable[match(Features(objTE_soloTE),
  ↪conversionTable$soloteID), "class"]])
soloTEageFamilyOld <-
  ↪cbind(soloTEageFamily["old",], colnames(soloTEageFamily), "old", "SoloTE")
colnames(soloTEageFamilyOld) <- c("nTEs", "superfamily", "age", "tool")
soloTEageFamilyYoung <-
  ↪cbind(soloTEageFamily["young",], colnames(soloTEageFamily), "young", "SoloTE")
colnames(soloTEageFamilyYoung) <- c("nTEs", "superfamily", "age", "tool")
soloTEageFamily <- rbind(soloTEageFamilyOld, soloTEageFamilyYoung)
soloTEageFamily <- as.data.frame(soloTEageFamily)
soloTEageFamily$nTEs <- as.numeric(soloTEageFamily$nTEs)

stellarscopeageFamily <-
  ↪table(conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "ageClass"],
  ↪conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "class"]])
stellarscopeageFamilyOld <-
  ↪cbind(stellarscopeageFamily["old",], colnames(stellarscopeageFamily), "old", "Stellarscope")
colnames(stellarscopeageFamilyOld) <- c("nTEs", "superfamily", "age", "tool")
stellarscopeageFamilyYoung <-
  ↪cbind(stellarscopeageFamily["young",], colnames(stellarscopeageFamily), "young", "Stellarscope")
colnames(stellarscopeageFamilyYoung) <- c("nTEs", "superfamily", "age", "tool")
stellarscopeageFamily <- rbind(stellarscopeageFamilyOld,
  ↪stellarscopeageFamilyYoung)
stellarscopeageFamily <- as.data.frame(stellarscopeageFamily)
stellarscopeageFamily$nTEs <- as.numeric(stellarscopeageFamily$nTEs)

ageFamilyDf <- rbind(soloTEageFamily, stellarscopeageFamily)
ageFamilyDf$superfamily <- gsub(ageFamilyDf$superfamily, pattern="\\?",
  ↪replacement="")

ageFamilyDf <- ageFamilyDf[ageFamilyDf$nTEs>0,]

ggplot(ageFamilyDf, aes(x=tool, fill=superfamily, y=nTEs)) +
  facet_wrap(~age) +
  geom_col(position="fill") +
  ↪#scale_fill_manual(values=adjustcolor(c("#5E50A1", "#3787BC", "#67C1A3", "#AADCA2", "#E6F596"),
  ↪alpha.f = 0.8)) +
  scale_fill_brewer(palette = "Spectral", direction = -1)+

```

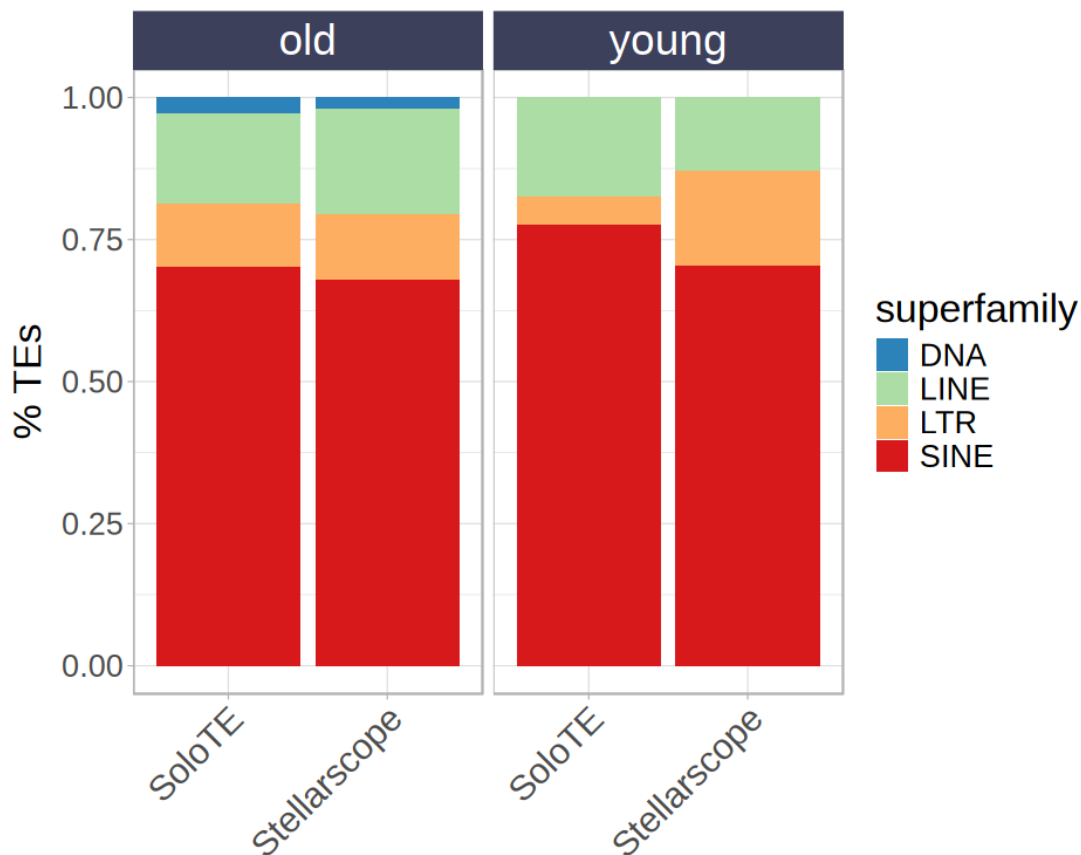
```

#palette = "Spectral", direction = -1)+
# scale_color_manual(values=c("#81B29A", "#9181A6"))+
ggtitle("") +
xlab("") +
ylab("% TEs") +
theme_light() +
theme(text=element_text(size=20),
      strip.text.x = element_text(size=22, hjust=0.5),
      strip.background = element_rect(fill="#3d405b", #"#e78063"
      axis.text.x = element_text(size=18, angle=45, hjust=1))

ggsave(paste0("figures_", dataset_id, "/
↳ nTEs_SoloTEvsStellarscope_barplot_age_order.pdf"), device = "pdf")

```

Saving 6.67 x 6.67 in image



```

[59]: options(repr.plot.width=4, repr.plot.height=6)

nOldSoloTE <- table(conversionTable[match(Features(objTE_soloTE),
  ↪conversionTable$soloteID), "ageClass"])[ "old"]
nYoungSoloTE <- table(conversionTable[match(Features(objTE_soloTE),
  ↪conversionTable$soloteID), "ageClass"])[ "young"]
nOldStellarscope <- table(conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "ageClass"])[ "old"]
nYoungStellarscope <- table(conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "ageClass"])[ "young"]
nOldSTARsolo <- table(conversionTable[match(Features(objTE_STARsolo),
  ↪conversionTable$stellarscopeID), "ageClass"])[ "old"]
nYoungSTARsolo <- table(conversionTable[match(Features(objTE_STARsolo),
  ↪conversionTable$stellarscopeID), "ageClass"])[ "young"]

table(conversionTable[match(Features(objTE_stellarscope),
  ↪conversionTable$stellarscopeID), "ageClass"])

df <- rbind( c(nOldSoloTE , nOldSoloTE / (nOldSoloTE + nYoungSoloTE), "old",
  ↪"SoloTE"),
             c(nYoungSoloTE, nYoungSoloTE / (nOldSoloTE + nYoungSoloTE),
  ↪"young", "SoloTE"),
             c(nOldStellarscope, nOldStellarscope / (nOldStellarscope +
  ↪nYoungStellarscope), "old", "Stellarscope"),
             c(nYoungStellarscope, nYoungStellarscope / (nOldStellarscope +
  ↪nYoungStellarscope), "young", "Stellarscope"),
             c(nOldSTARsolo, nOldSTARsolo / (nOldSTARsolo + nYoungSTARsolo),
  ↪"old", "STARsolo"),
             c(nYoungSTARsolo, nYoungSTARsolo / (nOldSTARsolo +
  ↪nYoungSTARsolo), "young", "STARsolo")
           )
df <- as.data.frame(df)

colnames(df) <- c("n_TEs", "n_TEs_prop", "AgeClass", "Tool")
df$n_TEs <- as.numeric(df$n_TEs)
df$n_TEs_prop <- as.numeric(df$n_TEs_prop)
df$n_TEs_perc <- as.numeric(df$n_TEs_prop) * 100

ggplot(df[df$AgeClass=="young",], aes(x=Tool, fill=Tool, color=Tool,
  ↪y=n_TEs_perc)) +
  geom_col(position = "stack") +
  scale_fill_manual(values=colorTools)+

```

```

scale_color_manual(values=colorTools)+
ggtitle("") +
xlab("") +
ylab("% young TEs") +
theme_minimal() +
theme(text=element_text(size=18),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.x = element_text(size=18, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/barplot_percYoungTEs.pdf"), device = "pdf",
      width = 4, height=6)

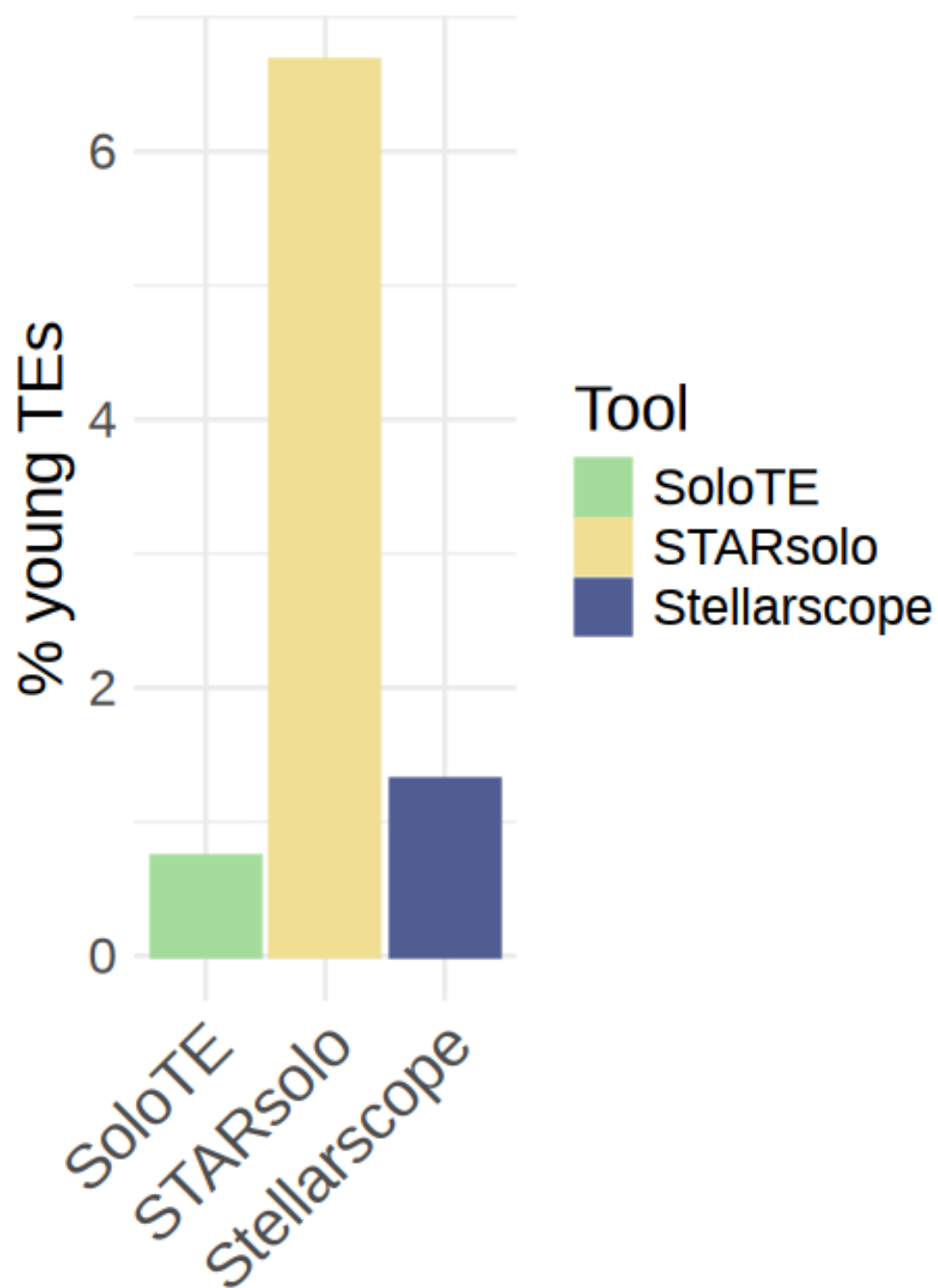
ggplot(df[df$AgeClass=="young",], aes(x=Tool, fill=Tool, color=Tool, y=n_TEs))
+
geom_col(position = "stack") +
scale_fill_manual(values=colorTools)+
scale_color_manual(values=colorTools)+
ggtitle("") +
xlab("") +
ylab("N. young TEs") +
theme_minimal() +
theme(text=element_text(size=18),
      plot.title = element_text(size=18, hjust=0.5),
      plot.subtitle = element_text(size=18, hjust=0.5),
      axis.text.x = element_text(size=18, angle=45, hjust=1))
ggsave(paste0("figures_",dataset_id,"/barplot_nYoungTEs.pdf"), device = "pdf",
      width = 4, height=6)

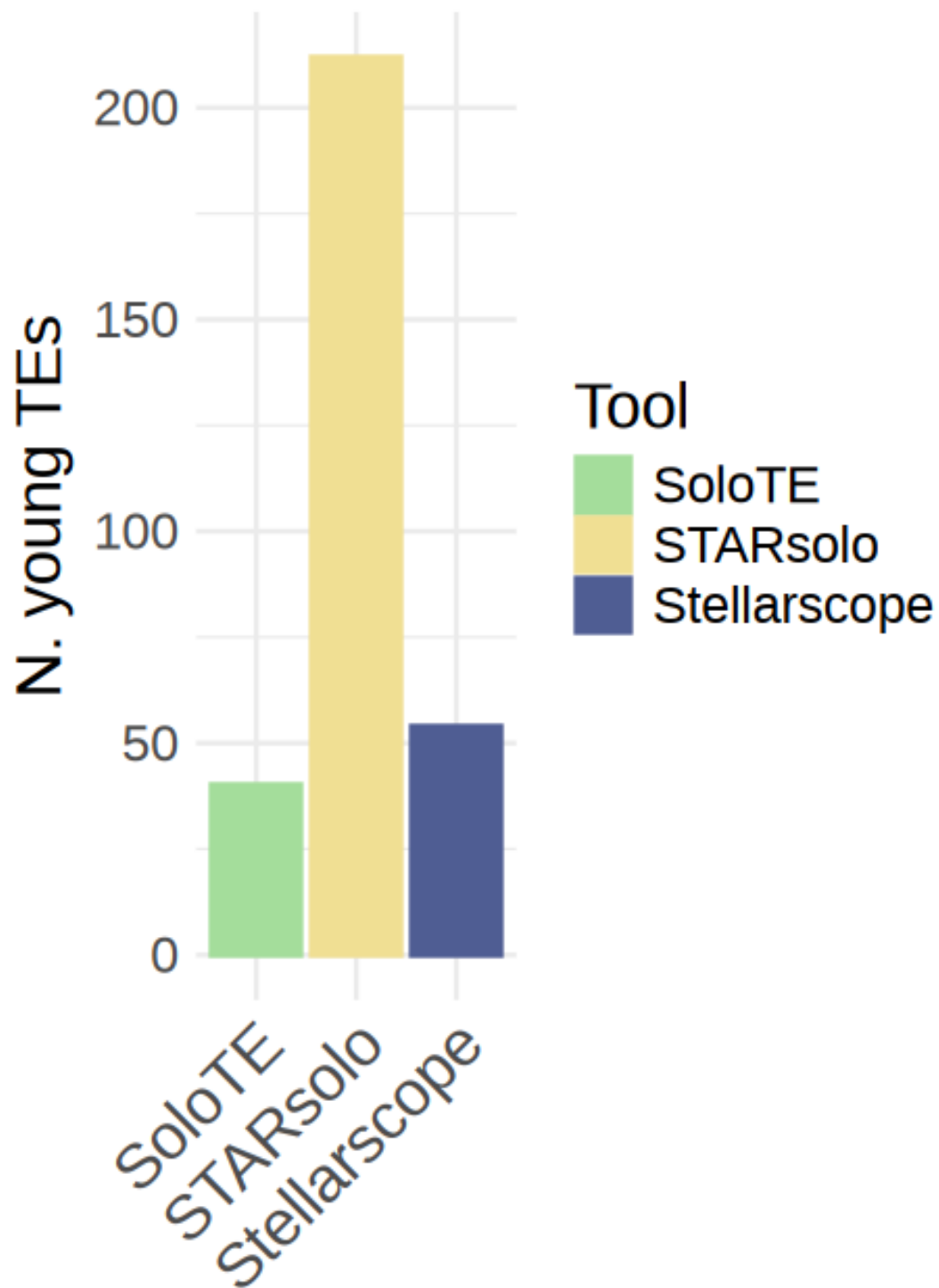
```

```

old young
4062    54

```





6 Celltype purity

```
[4]: celltypes <- read.table(paste0("data_", dataset_id, "/celltype_annotation.tsv"),  
  ↪ sep = "\t")  
colnames(celltypes) <- c("barcode", "celltype")  
rownames(celltypes) <- celltypes$barcode  
  
PBMCelltypeColors <- c("B_cells"="#6D5A5D",  
  "Dendritic Cells"="#C4B3AB", "#E78063", "#c492a7",  
  "Monocytes"="#81B29A",  
  "Platelets"="#e78063",  
  "T_cells"="#84B6D6",  
  "Natural Killers" = "#F2CC8F",  
  "Neutrophils"="#c492a7",  
  "Unknown" = "#3D405B")  
  
[5]: objTE_stellarscope <- FindNeighbors(objTE_stellarscope, dims = 1:10, k.param =  
  ↪ 20, return.neighbor = T)  
  
purity <- sapply(Cells(objTE_stellarscope), FUN = function(cell){  
  n_cells <- TopNeighbors(objTE_stellarscope@neighbors$RNA.nn, cell, n = 20)  
  celltypes <- objTE_stellarscope@meta.data[n_cells, 'celltype']  
  label <- objTE_stellarscope@meta.data[cell, 'celltype']  
  perc_label <- length(celltypes[celltypes==label]) / 20  
})  
df <- as.data.frame(cbind(purity, "Stellarscope"))  
colnames(df) <- c("purity", "tool")  
df$celltype <- objTE_stellarscope@meta.data[rownames(df), 'celltype']  
df$purity <- as.numeric(df$purity)  
  
objTE_soloTE <- FindNeighbors(objTE_soloTE, dims = 1:10, k.param = 20, return.  
  ↪ neighbor = T)  
objTE_soloTE$celltype <- celltypes$celltype[match(Cells(objTE_soloTE),  
  ↪ celltypes$barcode)]  
  
purity <- sapply(Cells(objTE_soloTE), FUN = function(cell){  
  n_cells <- TopNeighbors(objTE_soloTE@neighbors$RNA.nn, cell, n = 20)  
  celltypes <- objTE_soloTE@meta.data[n_cells, 'celltype']  
  label <- objTE_soloTE@meta.data[cell, 'celltype']  
  perc_label <- length(celltypes[celltypes==label]) / 20  
})
```



```

df2 <- as.data.frame(cbind(purity, "SoloTE"))
colnames(df2) <- c("purity", "tool")
df2$celltype <- objTE_soloTE@meta.data[rownames(df2), 'celltype']
df2$purity <- as.numeric(df2$purity)

objTE_STARsolo_sameCells <- objTE_STARsolo[, Cells(objTE_soloTE)]
objTE_STARsolo_sameCells <- FindNeighbors(objTE_STARsolo_sameCells, dims = 1:
  ↪10, k.param = 20, return.neighbor = T)

purity <- sapply(Cells(objTE_STARsolo_sameCells), FUN = function(cell){
  n_cells <- TopNeighbors(objTE_STARsolo_sameCells@neighbors$RNA.nn, cell, n = 20)
  ↪20)
  celltypes <- objTE_STARsolo_sameCells@meta.data[n_cells, 'celltype']
  label <- objTE_STARsolo_sameCells@meta.data[cell, 'celltype']
  perc_label <- length(celltypes[celltypes==label]) / 20
})
df3 <- as.data.frame(cbind(purity, "STARsolo"))
colnames(df3) <- c("purity", "tool")
df3$celltype <- objTE_STARsolo_sameCells@meta.data[rownames(df2), 'celltype']
df3$purity <- as.numeric(df3$purity)

df_purity <- rbind(df, df2, df3)

```

Computing nearest neighbors

Only one graph name supplied, storing nearest-neighbor graph only

Computing nearest neighbors

Only one graph name supplied, storing nearest-neighbor graph only

Computing nearest neighbors

Only one graph name supplied, storing nearest-neighbor graph only

```

[8]: # colorTools = c("SoloTE"="#A4DD9B", #6fc69d",
#               "Stellarscope"="#4f5d93",
#               "STARsolo"="#f0df93")
df_purity$tool <- factor(df_purity$tool, levels = c(
  ↪c("STARsolo", "Stellarscope", "SoloTE"))

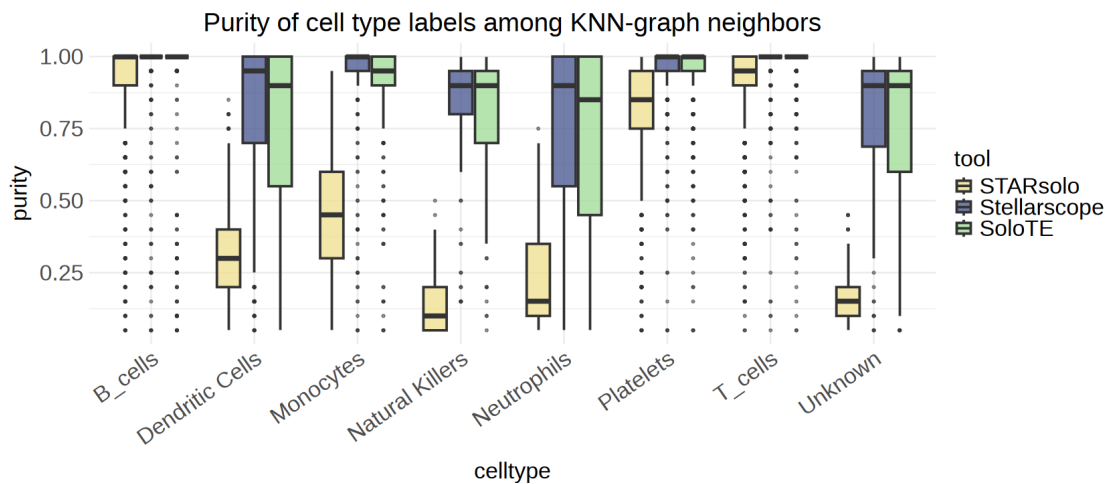
options(repr.plot.width=14, repr.plot.height=6)

```

```

ggplot(df_purity) + aes(y=purity, x=celltype, fill=tool) +
  geom_boxplot(linewidth = 0.8, alpha=0.8, outlier.size=0.7, outlier.alpha = 0.
  5) +
  ggtitle("Purity of cell type labels among KNN-graph neighbors") +
  scale_fill_manual(values=colorTools)+
  theme_minimal() +
  theme(text=element_text(size=20),
        plot.title = element_text(size=23, hjust=0.5),
        legend.text = element_text(size=20),
        axis.text.x = element_text(size=20, angle=35, hjust=1),
        axis.text.y = element_text(size=20, hjust=1))
ggsave(paste0("figures_", dataset_id, "/"
  celltype_purity_knn_soloTEvsStellarscopevsSTAR_noPlatelet.pdf"), width=14,
  height=6 )

```



[]: